

Here the right hand limit, i.e.,

$$\begin{aligned} f(0+0) &= \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \frac{e^{1/h} - 1}{e^{1/h} + 1} \\ &= \lim_{h \rightarrow 0} \frac{e^{1/h} [1 - (1/e^{1/h})]}{e^{1/h} [1 + (1/e^{1/h})]} = 1. \end{aligned}$$

Again the left hand limit, i.e.,

$$\begin{aligned} f(0-0) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} \frac{e^{-1/h} - 1}{e^{-1/h} + 1} \\ &= \lim_{h \rightarrow 0} \frac{(1/e^{1/h}) - 1}{(1/e^{1/h}) + 1} = \frac{0-1}{0+1} = -1. \end{aligned}$$

Since  $f(0+0) \neq f(0-0)$ , hence  $\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$  does not exist.

(iii) We have the right hand limit i.e.,  $f(0+0)$

$$\begin{aligned} &= \lim_{h \rightarrow 0} f(0+h), \text{ where } h \text{ is +ve but sufficiently small} \\ &= \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} h, \quad [\because h > 0 \text{ and } f(x) = x \text{ if } x > 0] \\ &= 0. \end{aligned}$$

Also, the left hand limit, i.e.,  $f(0-0)$

$$\begin{aligned} &= \lim_{h \rightarrow 0} f(0-h), \text{ where } h \text{ is +ve but sufficiently small} \\ &= \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} -(-h), \quad [\because h < 0 \text{ and } f(x) = -x \text{ if } x < 0] \\ &= \lim_{h \rightarrow 0} h = 0. \end{aligned}$$

Thus both the limits  $f(0+0)$  and  $f(0-0)$  exist and are equal to zero.

Hence  $\lim_{x \rightarrow 0} f(x)$  exists and is equal to zero.

Ex. 9. Let  $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ -x & \text{if } x \text{ is irrational} \end{cases}$

Show that  $\lim_{x \rightarrow a} f(x)$  exists only when  $a = 0$ .

Sol. Case I. If  $a$  is a non-zero rational number.

In this case  $f(a-0) = \lim_{h \rightarrow 0} f(a-h)$

$$= \lim_{h \rightarrow 0} (a-h) \quad \text{or} \quad \lim_{h \rightarrow 0} -(a-h),$$

according as  $(a-h)$  is rational or irrational

$= a$  or  $-a$  i.e., is not unique.

$\therefore f(a-0)$  does not exist.

$\therefore \lim_{x \rightarrow a} f(x)$  does not exist.

Case II. If  $a = 0$ . In this case  $f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h)$

$$= \lim_{h \rightarrow 0} (-h) \quad \text{or} \quad \lim_{h \rightarrow 0} h, \text{ according as } -h \text{ is rational or irrational}$$

$$= 0.$$

$$\text{Again } f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h)$$

$$= \lim_{h \rightarrow 0} h \quad \text{or} \quad \lim_{h \rightarrow 0} (-h), \text{ according as } h \text{ is rational or irrational}$$

$$= 0.$$

Since  $f(0+0) = f(0-0) = 0$ , hence  $\lim_{x \rightarrow 0} f(x)$  exists and is equal to zero.

Case III. If  $a$  is an irrational number.

In this case  $f(a-0) = \lim_{h \rightarrow 0} f(a-h)$

$$= \lim_{h \rightarrow 0} (a-h) \quad \text{or} \quad \lim_{h \rightarrow 0} -(a-h),$$

according as  $(a-h)$  is rational or irrational

$= a$  or  $-a$  i.e., is not unique.

$\therefore f(a-0)$  does not exist.

$\therefore \lim_{x \rightarrow a} f(x)$  does not exist.

Thus we see that  $\lim_{x \rightarrow a} f(x)$  exists only when  $a = 0$ .

Ex. 10. Discuss the existence of the limit of the function  $f$  defined as

$f(x) = 1$ , if  $x < 1$ ;  $f(x) = 2-x$ , if  $1 < x < 2$ ;  $f(x) = 2$ , if  $x \geq 2$

at  $x = 1$  and  $x = 2$ .

Sol. At  $x = 1$ . We have

$$\begin{aligned} f(1+0) &= \lim_{h \rightarrow 0} f(1+h), \text{ where } h \text{ is +ve and sufficiently small} \\ &= \lim_{h \rightarrow 0} [2 - (1+h)] = \lim_{h \rightarrow 0} (1-h) = 1; \end{aligned}$$

$$\text{and } f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} (1) = 1.$$

Since  $f(1+0) = f(1-0) = 1$ , hence  $\lim_{x \rightarrow 1} f(x)$  exists and is equal to 1.

At  $x = 2$ . We have  $f(2+0) = \lim_{h \rightarrow 0} f(2+h) = \lim_{h \rightarrow 0} (2) = 2$ ;

$$\text{and } f(2-0) = \lim_{h \rightarrow 0} f(2-h) = \lim_{h \rightarrow 0} [2 - (2-h)] = \lim_{h \rightarrow 0} h = 0.$$

Since  $f(2+0) \neq f(2-0)$ , hence  $\lim_{x \rightarrow 2} f(x)$  does not exist.

Ex. 11. If  $\lim_{x \rightarrow a} f(x) = \pm \infty$ , then  $\lim_{x \rightarrow a} \frac{1}{f(x)} = 0$ .

Sol. Let  $\lim_{x \rightarrow a} f(x) = +\infty$ .

Let  $\epsilon > 0$  be given. If  $\epsilon_1 = 1/\epsilon$ , then  $\epsilon_1 > 0$ .

Since  $\lim_{x \rightarrow a} f(x) = \infty$ , therefore for  $\varepsilon_1 > 0$ , there exists  $\delta > 0$  such that

$$\begin{aligned} f(x) > \varepsilon_1 & \text{ whenever } 0 < |x - a| < \delta \\ \text{i.e., } \frac{1}{f(x)} < \frac{1}{\varepsilon_1} & \text{ whenever } 0 < |x - a| < \delta \\ \text{i.e., } 0 < \frac{1}{f(x)} < \varepsilon & \text{ whenever } 0 < |x - a| < \delta \quad [\because \varepsilon = 1/\varepsilon_1] \\ \text{i.e., } -\varepsilon < \frac{1}{f(x)} < \varepsilon & \text{ whenever } 0 < |x - a| < \delta \\ \text{i.e., } \left| \frac{1}{f(x)} - 0 \right| < \varepsilon & \text{ whenever } 0 < |x - a| < \delta. \end{aligned}$$

$$\therefore \lim_{x \rightarrow a} \frac{1}{f(x)} = 0.$$

Similarly it can be proved that

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = 0 \text{ when } \lim_{x \rightarrow a} f(x) = -\infty.$$

Ex. 12. If  $f(x) = \frac{\sin [x]}{[x]}$ ,  $[x] \neq 0$  and  $f(x) = 0$ ,  $[x] = 0$ ,

where  $[x]$  denotes the greatest integer less than or equal to  $x$ , then find  $\lim_{x \rightarrow 0} f(x)$ .

(Kanpur 2010)

$$\begin{aligned} \text{Sol. Here } f(0+0) &= \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h) \\ &= \lim_{h \rightarrow 0} 0 \\ &= 0. \end{aligned}$$

$$[\because [h] = 0]$$

$$\text{Also } f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h)$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{\sin [-h]}{[-h]} \\ &= \lim_{h \rightarrow 0} \frac{\sin (-1)}{(-1)} = \frac{\sin (-1)}{(-1)} = \sin 1 \neq 0. \end{aligned}$$

$$[\because [-h] = -1 \neq 0]$$

Since  $f(0+0) \neq f(0-0)$ , therefore  $\lim_{x \rightarrow 0} f(x)$  does not exist.

### EXERCISE 1

- If  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  do not exist, can  $\lim_{x \rightarrow a} [f(x) + g(x)]$  or  $\lim_{x \rightarrow a} [f(x)g(x)]$  exist?
- If  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} [f(x) + g(x)]$  both exist, must  $\lim_{x \rightarrow a} g(x)$  exist?
- If  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} [f(x)g(x)]$  both exist, must  $\lim_{x \rightarrow a} g(x)$  exist?
- Show that  $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow a} f(x-a)$ .

- Using definition of limit, show that  $\lim_{x \rightarrow 0} f(x) = 1$  where

$$f(x) = \begin{cases} 1 + x^2 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

- If  $f$  is defined on  $\mathbb{R}$  as  $f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ 1, & \text{if } x \text{ is rational.} \end{cases}$  prove that  $\lim_{x \rightarrow a} f(x)$  does not exist for any  $a \in \mathbb{R}$ .
- If  $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$ , then prove that  $\lim_{x \rightarrow a} f(x) = f(a)$ .
- If  $x \rightarrow 0$ , then does the limit of the following function  $f$  exist or not?  
 $f(x) = x$ , when  $x < 0$ ;  $f(x) = 1$ , when  $x = 0$ ;  $f(x) = x^2$ , when  $x > 0$ .
- Use the formula  $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$  to find  $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$ .
- If  $f(x) = e^{-1/x}$ , show that at  $x = 0$ , the right hand limit is zero while the left hand limit is  $+\infty$ , and thus there is no limit of the function at  $x = 0$ .
- Prove that  $\lim_{x \rightarrow 0+0} f(x) = \lim_{x \rightarrow 0-0} f(-x)$ .
- Give an example to show that  $\lim_{x \rightarrow a} f(x)$  may exist even when the function is not defined for  $x = a$ .
- Let  $f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 3-x, & 1 \leq x \leq 2. \end{cases}$  Show that  $\lim_{x \rightarrow 1+} f(x) = 2$ . Does the limit of  $f(x)$  at  $x = 1$  exist? Give reasons for your answer.
- Evaluate  $\lim_{x \rightarrow 0} \frac{x - |x|}{x}$ .
- Evaluate  $\lim_{x \rightarrow 0} \frac{|\sin x|}{x}$ .
- Evaluate  $\lim_{x \rightarrow 0} \frac{e^{1/x}}{e^{1/x} + 1}$ .
- If  $f$  is defined on  $\mathbb{R}$  as  $f(x) = \begin{cases} 2, & \text{if } x \text{ is irrational} \\ 1, & \text{if } x \text{ is rational.} \end{cases}$  prove that  $\lim_{x \rightarrow a} f(x)$  does not exist for any  $a \in \mathbb{R}$ .

### ANSWERS

- Yes. If we take  $f(x) = \sin(1/x)$ ,  $g(x) = -\sin(1/x)$  whenever  $x \neq 0$ , then  $\lim_{x \rightarrow 0} f(x)$  and  $\lim_{x \rightarrow 0} g(x)$  do not exist but  $\lim_{x \rightarrow 0} [f(x) + g(x)]$  exists. Again if we take  $f(x) = g(x) = 1$  for all rational

$x$  and  $f(x) = g(x) = -1$  for all irrational  $x$ , then both  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  do not exist for any real number  $a$  but  $\lim_{x \rightarrow a} [f(x)g(x)]$  exists for every real number  $a$ .

2. Yes.

3. No. If we take  $f(x) = x \forall x \in \mathbb{R}$  and  $g(x) = \sin(1/x)$ , if  $x \neq 0$ ,  $g(0) = 0$ , then  $\lim_{x \rightarrow 0} f(x)$  and  $\lim_{x \rightarrow 0} [f(x)g(x)]$  both exist

but  $\lim_{x \rightarrow 0} g(x)$  does not exist.

13. Does not exist.

8. Yes:  $\lim_{x \rightarrow 0} f(x) = 0$ . 9.  $2 \log 2$ .

14. Right hand limit is 0 and left hand limit is 2 and so the limit does not exist.

15. Does not exist because the right hand limit is 1 and the left hand limit is -1.

16. The limit does not exist because the right hand limit is 1 and the left hand limit is 0.

### § 7. CONTINUITY.

The intuitive concept of continuity is derived from geometrical considerations. If the graph of the function  $y = f(x)$  is a continuous curve, it is natural to call the function continuous. This requires that there should be no sudden changes in the value of the function. A small change in  $x$  should produce only a small change in  $y$ . Moreover for the graph to be a continuous running curve, it should possess a definite direction at each point.

But the continuity as defined in pure analysis is quite distinct from the intuitive or the geometrical concept of the term. Sometimes drawing a graph is difficult. We now give the arithmetical definition of continuity given by Cauchy.

**Cauchy's definition of continuity.** A real valued function  $f$  defined on an open interval  $I$  is said to be continuous at  $a \in I$  iff for any arbitrarily chosen positive number  $\epsilon$ , however small, we can find a corresponding number  $\delta > 0$  such that

$$|f(x) - f(a)| < \epsilon \text{ whenever } |x - a| < \delta. \quad \dots(1)$$

(Bundelkhand 2010)

We say that  $f$  is a **continuous function** if it is continuous at every  $x \in I$ .

In other words,  $f$  is continuous at  $a$  if for any given  $\epsilon > 0$ , we can find a  $\delta > 0$  such that

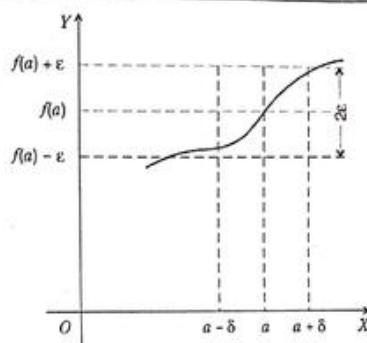
$$|x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon.$$

This means that the function  $f$  will be continuous at  $x = a$  if the difference between  $f(a)$  and the value of  $f(x)$  at any point in the interval  $|a - \delta, a + \delta|$  can be made less than a pre-assigned positive number  $\epsilon$ . Note that we choose  $\delta$  after we have chosen  $\epsilon$ .

A geometrical interpretation of the above definition is immediate. Corresponding to any pre-assigned positive number  $\epsilon$ , we can determine an interval of width  $2\delta$  about the point  $x = a$  (see the figure) such that for any point  $x$  lying in the interval  $|a - \delta, a + \delta|$ ,  $f(x)$  is confined to lie between  $f(a) - \epsilon$  and  $f(a) + \epsilon$ .

The inequality (1) may be written in the form of an equality as

$$f(x) = f(a) + \eta, \text{ where } |\eta| < \epsilon.$$



**Note 1.** For a function  $f(x)$  to be continuous at  $x = a$ , it is necessary that  $\lim_{x \rightarrow a} f(x)$  must exist.

**Note 2.** The function must be defined at the point of continuity.

**Note 3.** The value of  $\delta$  depends upon the values of  $\epsilon$  and  $a$ .

**Note 4.** The interval  $I$  may be of any one of the forms:

$$]a, b[, ] - \infty, b[, ]a, \infty[, ] - \infty, \infty[.$$

**An alternative definition of continuity.** A function  $f$  is said to be continuous at  $a \in I$  iff  $\lim_{x \rightarrow a} f(x)$  exists, is finite and is equal to  $f(a)$  otherwise the function is discontinuous at  $x = a$ .

This definition of continuity follows immediately from the definition of limit and the definition of continuity. Thus a function  $f$  is said to be continuous at  $a$ , if  $f(a+0) = f(a-0) = f(a)$ . This is a working formula for testing the continuity of a function at a given point. (Bundelkhand 2008)

**Important Remark.** If  $f(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n$  is a polynomial in  $x$  of degree  $n$ , then by the above definition it can be easily seen that  $f(x)$  is continuous for all  $x \in \mathbb{R}$ .

If  $c$  be any real number, then

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \{a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n\} \\ &= a_0 \lim_{x \rightarrow c} x^n + a_1 \lim_{x \rightarrow c} x^{n-1} + \dots + a_{n-1} \lim_{x \rightarrow c} x + \lim_{x \rightarrow c} a_n \\ &= a_0c^n + a_1c^{n-1} + \dots + a_{n-1}c + a_n \quad \left[ \because \lim_{x \rightarrow c} x = c \right] \\ &= f(c). \end{aligned}$$

Since  $\lim_{x \rightarrow c} f(x) = f(c)$ , therefore  $f(x)$  is continuous at  $x = c$ .

Thus  $f(x)$  is continuous at every real number  $c$  and so  $f(x)$  is continuous for all  $x \in \mathbb{R}$ .

Thus remember that a polynomial function  $f(x)$  is always continuous at each point of its domain.

### Continuity from left and continuity from right.

Let  $f$  be a function defined on an open interval  $I$  and let  $a \in I$ . We say that  $f$  is continuous from the left at  $a$  if  $\lim_{x \rightarrow a^-} f(x)$  exists and is equal to  $f(a)$ . Similarly  $f$  is said to be continuous from the right at  $a$  if  $\lim_{x \rightarrow a^+} f(x)$  exists and is equal to  $f(a)$ .

From the above definitions it is clear that for a function  $f$  to be continuous at  $a$ , it is necessary as well as sufficient that  $f$  be continuous from the left as well as from the right at  $a$ .

**Continuous function.** A function  $f$  is said to be a continuous function if it is continuous at each point of its domain.

**Continuity in an open interval.** A function  $f$  is said to be continuous in the open interval  $[a, b]$  if it is continuous at each point of the interval. (Bundelkhand 2009)

**Continuity in a closed interval.** Let  $f$  be a function defined on the closed interval  $[a, b]$ . We say that  $f$  is continuous at  $a$  if it is continuous from the right at  $a$  and also that  $f$  is continuous at  $b$  if it is continuous from the left at  $b$ . Further,  $f$  is said to be continuous on the closed interval  $[a, b]$ , if (i) it is continuous from the right at  $a$ , (ii) continuous from the left at  $b$  and (iii) continuous on the open interval  $(a, b)$ , then

Thus if a function  $f$  is defined on the closed interval  $[a, b]$ , then

(i) it is continuous at the left end point  $a$  if  $f(a) = f(a+0)$

$$\text{i.e., } f(a) = \lim_{x \rightarrow a+0} f(x)$$

(ii) it is continuous at the right end point  $b$  if  $f(b) = f(b-0)$

$$\text{i.e., } f(b) = \lim_{x \rightarrow b-0} f(x)$$

and (iii) it is continuous at an interior point  $c$  of  $[a, b]$  i.e., at  $c \in ]a, b[$  if

$$f(c-0) = f(c) = f(c+0) \text{ i.e., if } \lim_{x \rightarrow c-0} f(x) = f(c) = \lim_{x \rightarrow c+0} f(x).$$

### § 8. DISCONTINUITY.

**Definition.** If a function is not continuous at a point, then it is said to be discontinuous at that point and the point is called a point of discontinuity of this function.

#### Types of discontinuity.

(i) **Removable discontinuity.**

(Meerut 2011)

A function  $f$  is said to have a removable discontinuity at a point  $a$  if  $\lim_{x \rightarrow a} f(x)$  exists but is not equal to  $f(a)$  i.e., if

$$f(a+0) = f(a-0) \neq f(a).$$

The function can be made continuous by defining it in such a way that

$$\lim_{x \rightarrow a} f(x) = f(a).$$

(ii) **Discontinuity of the first kind or ordinary discontinuity.** (Meerut 2010B)

A function  $f$  is said to have a discontinuity of the first kind or ordinary discontinuity at  $a$  if  $f(a+0)$  and  $f(a-0)$  both exist but are not equal. The point  $a$  is said to be a point of discontinuity from the left or right according as  $f(a-0) \neq f(a) = f(a+0)$  or  $f(a+0) \neq f(a) = f(a-0)$ .

(iii) **Discontinuity of the second kind.** A function  $f$  is said to have a discontinuity of the second kind, at  $a$  if none of the limits  $f(a+0)$  and  $f(a-0)$  exist. The point  $a$  is said to be a point of discontinuity of the second kind from the left or right according as  $f(a-0)$  or  $f(a+0)$  does not exist.

(Meerut 2003, 10B)

(iv) **Mixed discontinuity.** A function  $f$  is said to have a mixed discontinuity at  $a$ , if  $f$  has a discontinuity of second kind on one side of  $a$  and on the other side a discontinuity of first kind or may be continuous.

(v) **Infinite discontinuity.** A function  $f$  is said to have an infinite discontinuity at  $a$  if  $f(a+0)$  or  $f(a-0)$  is  $+\infty$  or  $-\infty$ . Obviously, if  $f$  has a discontinuity at  $a$  and is unbounded in every neighbourhood of  $a$ , then  $f$  is said to have an infinite discontinuity at  $a$ .

### § 9. JUMP OF A FUNCTION AT A POINT.

If both  $f(a+0)$  and  $f(a-0)$  exist, then the jump in the function at  $a$  is defined as the non-negative difference  $f(a+0) - f(a-0)$ . A function having a finite number of jumps in a given interval is called piecewise continuous or sectionally continuous.

### SOLVED EXAMPLES

**Ex. 1.** Test the following functions for continuity:

(i)  $f(x) = x \sin(1/x)$ ,  $x \neq 0$ ,  $f(0) = 0$  at  $x = 0$ .

Also draw the graph of the function.

(Meerut 2009B)

(ii)  $f(x) = 2^{1/x}$  when  $x \neq 0$ ,  $f(0) = 0$  at  $x = 0$ .

(iii)  $f(x) = 1/(1 - e^{-1/x})$ ,  $x \neq 0$ ,  $f(0) = 0$  at  $x = 0$ .

**Sol.** (i) Here  $f(0+0) = \lim_{h \rightarrow 0^+} f(0+h)$ ,  $h > 0$

$$= \lim_{h \rightarrow 0^+} f(h) = \lim_{h \rightarrow 0^+} h \sin \frac{1}{h} = 0.$$

[See theorem 10 of § 3]

$$\left[ \because \lim_{h \rightarrow 0} h = 0 \text{ and } \left| \sin \frac{1}{h} \right| \leq 1 \text{ for all } h \neq 0 \text{ i.e., } \sin(1/h) \text{ is bounded in some deleted neighbourhood of zero} \right]$$

Similarly  $f(0-0) = \lim_{h \rightarrow 0^-} f(0-h)$ ,  $h > 0$

$$= \lim_{h \rightarrow 0^-} f(-h) = \lim_{h \rightarrow 0^-} (-h) \sin \left( \frac{1}{-h} \right) = \lim_{h \rightarrow 0^-} h \sin \frac{1}{h} = 0, \text{ as before.}$$

Also  $f(0) = 0$ .

Thus  $f(0-0) = f(0) = f(0+0)$ .

$\therefore$  the function  $f(x)$  is continuous at  $x = 0$ .

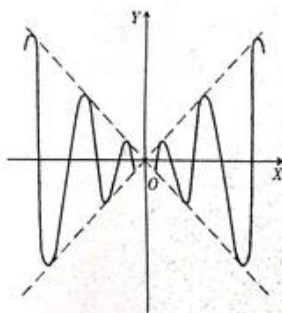
To draw the graph of the function we put  $y = f(x)$ .

So the graph of the function is the curve

$$y = x \sin(1/x), x \neq 0$$

$$y = 0 \text{ when } x = 0.$$

If we put  $-x$  in place of  $x$ , the equation of this curve does not change and so this curve is symmetrical about the  $y$ -axis and it is sufficient to draw the graph when  $x > 0$ .



$$\text{Also } |f(x)| = |x \sin(1/x)| = |x| \cdot |\sin(1/x)| \leq |x|$$

$\therefore$  for all  $x$  the curve  $y = x \sin(1/x)$  lies between the lines  $y = x$  and  $y = -x$ .  
Excluding origin the curve meets the  $y$ -axis at the points where

$$\sin \frac{1}{x} = 0 \text{ i.e., where } \frac{1}{x} = \pi, 2\pi, 3\pi, \dots \text{ i.e., where } x = \frac{1}{\pi}, \frac{1}{2\pi}, \frac{1}{3\pi}, \dots$$

$$\text{Also } y = x \text{ at the points where } \sin \frac{1}{x} = 1 \text{ i.e., } \frac{1}{x} = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \dots$$

$$\text{i.e., } x = \frac{2}{\pi}, \frac{2}{5\pi}, \frac{2}{9\pi}, \dots$$

$$\text{and } y = -x \text{ at the points where } \sin \frac{1}{x} = -1 \text{ i.e., } \frac{1}{x} = \frac{3\pi}{2}, \frac{7\pi}{2}, \dots$$

$$\text{i.e., } x = \frac{2}{3\pi}, \frac{2}{7\pi}, \dots$$

$$\text{We have } \frac{dy}{dx} = \sin \frac{1}{x} + x \left( \cos \frac{1}{x} \right) \left( -\frac{1}{x^2} \right) = \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x}$$

So at the points where  $\sin(1/x) = 1$ , we have  $\cos(1/x) = 0$  and  $dy/dx = 1$  i.e., at these points the curve touches the straight line  $y = x$ . Similarly at the points where  $\sin(1/x) = -1$ , the curve touches the straight line  $y = -x$ .

$$\begin{aligned} \text{Also } \lim_{x \rightarrow \infty} x \sin \frac{1}{x} & \quad [\text{Form } \infty \times 0] \\ &= \lim_{x \rightarrow \infty} \frac{\sin(1/x)}{1/x} \quad [\text{Form } \frac{0}{0}] \\ &= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \text{putting } \frac{1}{x} = \theta \text{ so that } \theta \rightarrow 0 \text{ as } x \rightarrow \infty \\ &= 1. \end{aligned}$$

Thus  $y \rightarrow 1$  as  $x \rightarrow \infty$  and so the straight line  $y = 1$  is an asymptote of the curve.

Although the function is continuous at the origin, yet the graph of the function in the vicinity of the origin cannot be drawn, since the function oscillates infinitely often in any interval containing the origin.

From the graph it is clear that the function makes an infinite number of oscillations in the neighbourhood of  $x = 0$ . The oscillations, however, go on diminishing in length as  $x \rightarrow 0$ .

**Note 1.** If we are to check the continuity of  $f(x)$  at any point  $x = c$ , where  $c \neq 0$ , then we see that  $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} x \sin \frac{1}{x} = c \sin \frac{1}{c} = f(c)$  and so  $f(x)$  is continuous at  $x = c$ .

Thus  $f(x)$  is continuous for all  $x \in \mathbb{R}$  i.e.,  $f(x)$  is continuous on the whole real line.

**Note 2.** If we take  $f(0) = 2$ , the function becomes discontinuous at  $x = 0$  and has a removable discontinuity at  $x = 0$ .

$$(ii) \text{ Here } f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} 2^{1/h} = 2^\infty = \infty,$$

$$f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} 2^{-1/h} = 2^{-\infty} = 0,$$

$$f(0) = 0.$$

Since  $f(0+0) \neq f(0-0)$ , therefore the function is discontinuous at the origin. It has an **infinite discontinuity** there.

$$(iii) \text{ Here } f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{1}{1 - e^{-1/h}} = 1,$$

$$f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{1}{1 - e^{1/h}} = 0.$$

Since  $f(0+0) \neq f(0-0)$ , hence  $f(x)$  is discontinuous at  $x = 0$  and has discontinuity of the first kind. This function has a jump of one unit at 0 since  $f(0+0) - f(0-0) = 1$ .

**Ex. 2.** Consider the function  $f$  defined by  $f(x) = x - [x]$ , where  $x$  is a positive variable and  $[x]$  denotes the integral part of  $x$  and show that it is discontinuous for integral values of  $x$  and continuous for all others. Draw its graph.

**Sol.** From the definition of the function  $f(x)$ , we have

$$f(x) = x - (n-1) \quad \text{for } n-1 < x < n,$$

$$f(x) = 0 \quad \text{for } x = n,$$

$$f(x) = x - n \quad \text{for } n < x < n+1, \text{ where } n \text{ is an integer.}$$

We shall test the function  $f(x)$  for continuity at  $x = n$ .

$$\text{We have } f(n) = 0;$$

$$\begin{aligned} f(n+0) &= \lim_{h \rightarrow 0} f(n+h) = \lim_{h \rightarrow 0} [(n+h) - n] \quad [\because n < n+h < n+1] \\ &= \lim_{h \rightarrow 0} h = 0; \end{aligned}$$

$$\begin{aligned} \text{and } f(n-0) &= \lim_{h \rightarrow 0} f(n-h) = \lim_{h \rightarrow 0} [(n-h) - (n-1)] \quad [\because n-1 < n-h < n] \\ &= \lim_{h \rightarrow 0} (1-h) = 1; \end{aligned}$$

Since  $f(n+0) \neq f(n-0)$ , the function  $f(x)$  is discontinuous at  $x = n$ . Thus  $f(x)$  is discontinuous for all integral values of  $x$ . It is obviously continuous for all other values of  $x$ .

Since  $x$  is a positive variable, putting  $n = 1, 2, 3, 4, 5, \dots$  we see that the graph of  $f(x)$  consists of the following straight lines:

$$y = x \text{ when } 0 < x < 1,$$

$$y = 0 \text{ when } x = 1$$

$$y = x-1 \text{ when } 1 < x < 2,$$

$$y = 0 \text{ when } x = 2$$

$$y = x-2 \text{ when } 2 < x < 3,$$

$$y = 0 \text{ when } x = 3$$

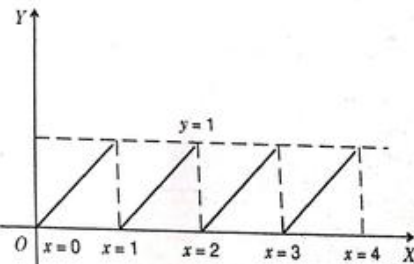
$$y = x-3 \text{ when } 3 < x < 4,$$

$$y = 0 \text{ when } x = 4 \text{ and so on.}$$

The graph of the function thus obtained is shown by thick lines from  $x = 0$  to  $x = 4$ . From the graph it is evident that:

(i) The function is discontinuous for all integral values of  $x$  but continuous for other values of  $x$ .

(ii) The function is bounded between 0 and 1 in every domain which includes an integer.



(iii) The lower bound 0 is attained but the upper bound 1 is not attained since  $f(x) \neq 1$  for any value of  $x$ .

**Ex. 3.** Show that the function  $f(x) = [x] + [-x]$  has removable discontinuity for integral values of  $x$ . (Kanpur 2009)

**Sol.** We observe that  $f(x) = 0$ , when  $x$  is an integer and  $f(x) = -1$ , when  $x$  is not an integer. Hence if  $n$  is any integer, we have  $f(n-0) = f(n+0) = -1$  and  $f(n) = 0$ . So the function  $f(x)$  has a removable discontinuity at  $x = n$ , where  $n$  is an integer.

**Ex. 4.** Let  $y = E(x)$ , where  $E(x)$  denotes the integral part of  $x$ . Prove that the function is discontinuous where  $x$  has an integral value. Also draw the graph.

**Sol.** From the definition of  $E(x)$ , we have

$$\begin{aligned} E(x) &= n-1 & \text{for } n-1 \leq x < n, \\ E(x) &= n & \text{for } n \leq x < n+1, \\ E(x) &= n+1 & \text{for } n+1 \leq x < n+2, \end{aligned}$$

and so on where  $n$  is an integer.

We consider  $x = n$ .

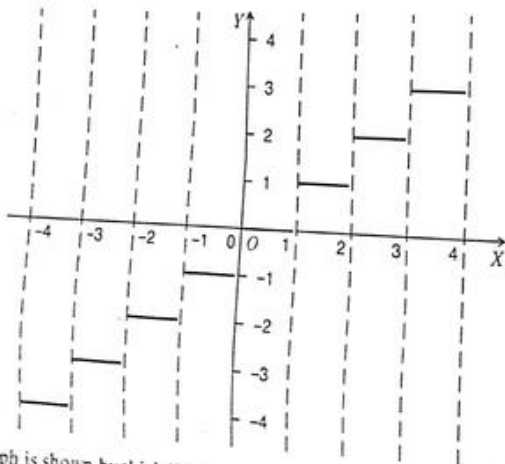
Then  $E(n) = n$ ,  $E(n-0) = n-1$  and  $E(n+0) = n$ .

Since  $E(n+0) \neq E(n-0)$ , the function  $E(x)$  is discontinuous at  $x = n$  i.e., when  $x$  has an integral value.

Evidently it is continuous for all other values of  $x$ .

To draw the graph, we put  $n = \dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots$ , so that

$$\begin{aligned} y &= -4, & \text{when } -4 \leq x < -3, \\ y &= -3, & \text{when } -3 \leq x < -2, \\ y &= -2, & \text{when } -2 \leq x < -1, \\ y &= -1, & \text{when } -1 \leq x < 0, \\ y &= 0, & \text{when } 0 \leq x < 1, \\ y &= 1, & \text{when } 1 \leq x < 2, \\ y &= 2, & \text{when } 2 \leq x < 3, \\ y &= 3, & \text{when } 3 \leq x < 4, \\ y &= 4, & \text{when } 4 \leq x < 5 \text{ and so on.} \end{aligned}$$



The graph is shown by thick lines.

## LIMITS AND CONTINUITY

**Ex. 5.** Show that the function  $\phi$  defined as

$$\phi(x) = \begin{cases} 0 & \text{for } x = 0 \\ \frac{1}{2} - x & \text{for } 0 < x < \frac{1}{2} \\ \frac{1}{2} & \text{for } x = \frac{1}{2} \\ \frac{3}{2} - x & \text{for } \frac{1}{2} < x < 1 \\ 1 & \text{for } x = 1 \end{cases}$$

has three points of discontinuity which you are required to find. Also draw the graph of the function.

**Sol.** Here the domain of the function  $\phi(x)$  is the closed interval  $[0, 1]$ .

When  $0 < x < \frac{1}{2}$ ,  $\phi(x) = \frac{1}{2} - x$  which is a polynomial in  $x$  of degree 1. We know that a polynomial function is continuous at each point of its domain and so  $\phi(x)$  is continuous at each point of the open interval  $0 < x < \frac{1}{2}$ .

Again when  $\frac{1}{2} < x < 1$ ,  $\phi(x) = \frac{3}{2} - x$  which is also a polynomial in  $x$  and so  $\phi(x)$  is also continuous at each point of the open interval  $\frac{1}{2} < x < 1$ .

Now it remains to test the function  $\phi(x)$  for continuity at  $x = 0$ ,  $\frac{1}{2}$  and 1.

(i) For  $x = 0$ , we have  $\phi(0) = 0$ ,

$$\phi(0+0) = \lim_{h \rightarrow 0} \phi(0+h) = \lim_{h \rightarrow 0} \phi(h) = \lim_{h \rightarrow 0} \left( \frac{1}{2} - h \right) = \frac{1}{2}.$$

Since  $\phi(0) \neq \phi(0+0)$ , the function  $\phi(x)$  is discontinuous at  $x = 0$  and the discontinuity is ordinary.

(ii) For  $x = \frac{1}{2}$ , we have  $\phi\left(\frac{1}{2}\right) = \frac{1}{2}$ ,

$$\begin{aligned} \phi\left(\frac{1}{2}-0\right) &= \lim_{h \rightarrow 0} \phi\left(\frac{1}{2}-h\right) = \lim_{h \rightarrow 0} \left[ \frac{1}{2} - \left(\frac{1}{2}-h\right) \right], \quad \left[ \text{Note that } 0 < \frac{1}{2}-h < \frac{1}{2} \right] \\ &= \lim_{h \rightarrow 0} h = 0. \end{aligned}$$

Since  $\phi\left(\frac{1}{2}-0\right) \neq \phi\left(\frac{1}{2}\right)$ , the function  $\phi(x)$  is discontinuous from the left at  $x = \frac{1}{2}$ .

Again  $\phi\left(\frac{1}{2}+0\right) = \lim_{h \rightarrow 0} \phi\left(\frac{1}{2}+h\right)$ ,  $h > 0$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \left[ \frac{3}{2} - \left(\frac{1}{2}+h\right) \right] \quad \left[ \because \frac{1}{2} < \frac{1}{2}+h < 1 \right] \\ &= \lim_{h \rightarrow 0} (1-h) = 1 \neq \phi\left(\frac{1}{2}\right) = \frac{1}{2}. \end{aligned}$$

Thus the function  $\phi(x)$  is discontinuous from the right also at  $x = \frac{1}{2}$ .

In this way  $\phi(x)$  has discontinuity of the first kind i.e., ordinary discontinuity at  $x = \frac{1}{2}$  and the jump of the function at  $x = \frac{1}{2}$  is  $\phi\left(\frac{1}{2}+0\right) - \phi\left(\frac{1}{2}-0\right)$  i.e.,  $1 - 0$  i.e., 1.

(iii) For  $x = 1$ , we have  $\phi(1) = 1$ ,

$$\phi(1-0) = \lim_{h \rightarrow 0} \phi(1-h)$$

$$= \lim_{h \rightarrow 0} \left[ \frac{3}{2} - (1-h) \right], \quad \left[ \text{Note that } \frac{1}{2} < 1-h < 1 \right]$$

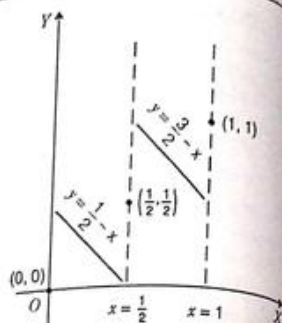
$$= \lim_{h \rightarrow 0} \left( \frac{1}{2} + h \right) = \frac{1}{2}$$

Since  $\phi(1) \neq \phi(1-0)$ ,  $\phi(x)$  is discontinuous at  $x = 1$  and the discontinuity is ordinary.

Hence the function  $\phi(x)$  has three points of discontinuity at  $x = 0, \frac{1}{2}$  and  $1$ .

The graph of the function consists of the point  $(0, 0)$ ; the segment of the line  $y = \frac{1}{2} - x$ ,  $0 < x < \frac{1}{2}$ ; the point  $(\frac{1}{2}, \frac{1}{2})$ ; the segment of the line  $y = \frac{3}{2} - x$ ,  $\frac{1}{2} < x < 1$ ; and the point  $(1, 1)$ .

Thus the graph is as shown in the figure. From the graph we observe that the function is discontinuous at  $x = 0, \frac{1}{2}$  and  $1$ .



Ex. 6. Determine the values of  $a, b, c$  for which the function

$$f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$$

is continuous at  $x = 0$ .

$$\text{Sol. Here } f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{(h+bh^2)^{1/2} - h^{1/2}}{bh^{3/2}}$$

$$= \lim_{h \rightarrow 0} \frac{(1+bh)^{1/2} - 1}{bh} = \lim_{h \rightarrow 0} \frac{[1 + \frac{1}{2}bh + \dots] - 1}{bh} = \frac{1}{2},$$

which is independent of  $b$  and so  $b$  may have any real value except  $0$ .

$$\begin{aligned} \text{Again } f(0-0) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{\sin(a+1)(-h) + \sin(-h)}{(-h)} \\ &= \lim_{h \rightarrow 0} \frac{\sin(a+1)h + \sin h}{h} = \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{1}{2}a+1\right)h \cos(ah/2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin\{(a+2)/2\}h}{\{(a+2)/2\}h} (a+2) \cos(ah/2) = a+2. \end{aligned}$$

For continuity at  $x = 0$ , we have  $f(0+0) = f(0-0) = f(0)$  i.e.,  $\frac{1}{2} = a+2 = c$ .  $\therefore c = \frac{1}{2}$  and  $a = -\frac{3}{2}$ .

Ex. 7. A function  $f(x)$  is defined as follows:

$$f(x) = \begin{cases} (x^2/a) - a, & \text{when } x < a \\ 0, & \text{when } x = a \\ a - (a^2/x), & \text{when } x > a. \end{cases}$$

Prove that the function  $f(x)$  is continuous at  $x = a$ .

$$\text{Sol. We have } f(a+0) = \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} \left[ a - \frac{a^2}{(a+h)} \right]$$

$$[\because f(x) = a - (a^2/x) \text{ for } x > a]$$

(Bundelkhand 2007)

$$= [a - (a^2/a)] = a - a = 0;$$

$$\begin{aligned} f(a-0) &= \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} \left[ \frac{(a-h)^2}{a} - a \right], [\because f(x) = (x^2/a) - a \text{ for } x < a] \\ &= [(a^2/a) - a] = a - a = 0. \end{aligned}$$

Also, we have  $f(a) = 0$ .

Since  $f(a+0) = f(a-0) = f(a)$ , therefore  $f(x)$  is continuous at  $x = a$ .

Ex. 8. Examine the function defined below for continuity at  $x = a$ :

$$f(x) = \frac{1}{x-a} \operatorname{cosec}\left(\frac{1}{x-a}\right), x \neq a$$

$$f(x) = 0, x = a.$$

$$\text{Sol. We have } f(a+0) = \lim_{h \rightarrow 0} f(a+h)$$

$$= \lim_{h \rightarrow 0} \frac{1}{a+h-a} \operatorname{cosec} \frac{1}{a+h-a} = \lim_{h \rightarrow 0} \frac{1}{h \sin(1/h)}$$

$$= +\infty, \text{ since } h \sin(1/h) \rightarrow 0 \text{ as } h \rightarrow 0.$$

$$f(a-0) = \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} \frac{1}{a-h-a} \operatorname{cosec}\left(\frac{1}{a-h-a}\right)$$

$$= \lim_{h \rightarrow 0} \left[ \frac{1}{h} \cdot \frac{1}{\sin\{-1/h\}} \right] = \lim_{h \rightarrow 0} \frac{1}{h \sin(1/h)}$$

$$= +\infty, \text{ since } h \sin(1/h) \rightarrow 0 \text{ as } h \rightarrow 0.$$

Also, we have  $f(a) = 0$ .

Since  $f(a+0) = f(a-0) \neq f(a)$ , the function  $f(x)$  is discontinuous at  $x = a$ , having an infinite discontinuity of the second kind.

Ex. 9. Examine the function defined below for continuity at  $x = 0$ :

$$f(x) = \frac{\sin^2 ax}{x^2} \text{ for } x \neq 0, f(x) = 1 \text{ for } x = 0.$$

(Meerut 2010)

Sol. We have  $f(0) = 1$ ;

$$f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \frac{\sin^2 ah}{h^2}$$

$$= \lim_{h \rightarrow 0} \left( \frac{\sin ah}{ah} \right)^2 \cdot a^2 = 1 \cdot a^2 = a^2;$$

$$\text{and } f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} \frac{\sin^2(-ah)}{(-h)^2}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^2 ah}{h^2} = a^2.$$

Now  $f(x)$  is continuous at  $x = 0$  iff

$$f(0+0) = f(0-0) = f(0).$$

Hence  $f(x)$  is discontinuous at  $x = 0$  unless  $a = 1$ .

Ex. 10. A function  $f(x)$  is defined as follows:

$$f(x) = 1+x \text{ if } x \leq 2 \text{ and } f(x) = 5-x \text{ if } x \geq 2.$$

Is the function continuous at  $x = 2$ ?

(Meerut 2002, 06)

Sol. Here  $f(2) = 1+2$  or  $5-2 = 3$ ;

$$f(2+0) = \lim_{h \rightarrow 0} f(2+h), \text{ where } h \text{ is +ive and sufficiently small}$$

Thus for any given  $\varepsilon > 0$ , there exists  $\delta > 0$  such that  $|f(x) - f(a)| < \varepsilon$ , whenever  $|x - a| < \delta$ .

$\therefore$  by definition of continuity of a function at a point, the function  $|f|$  is continuous at  $a$ .

Note: The converse is not true. For example, if

$$f(x) = -1, \text{ for } x < a \text{ and } f(x) = 1 \text{ for } x \geq a \text{ then}$$

$$\lim_{x \rightarrow a} |f(x)| = 1 = |f(a)|, \text{ but } \lim_{x \rightarrow a} f(x) \text{ does not exist.}$$

Thus  $|f|$  is continuous at  $a$  while  $f$  is not continuous at  $a$ .

### EXERCISE 2

1. Discuss the continuity and discontinuity of the following functions:

(i)  $f(x) = x^2 - 3x$

(ii)  $f(x) = x + x^{-1}$

(iii)  $f(x) = e^{-1/x}$

(iv)  $f(x) = \sin x$

(v)  $f(x) = \cos(1/x)$  when  $x \neq 0$ ,  $f(0) = 0$ .

(vi)  $f(x) = \sin(1/x)$  when  $x \neq 0$ , and  $f(0) = 0$ .

(vii)  $f(x) = \frac{\sin x}{x}$  when  $x \neq 0$  and  $f(0) = 1$ .

(viii)  $f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1}$  when  $x \neq 0$  and  $f(0) = 1$ .

(Meerut 2004B)

(ix)  $f(x) = \frac{e^{1/x} - 1}{1 + e^{1/x}}$  when  $x \neq 0$ ,  $f(0) = 0$ .

(Bundelkhand 2011)

(x)  $f(x) = \frac{x e^{1/x}}{1 + e^{1/x}} + \sin(1/x)$  when  $x \neq 0$ ,  $f(0) = 0$ .

(xi)  $f(x) = \sin x \cos(1/x)$  when  $x \neq 0$ ,  $f(0) = 0$ .

2. (i) Discuss the continuity of  $f(x)$  at  $x = 0$ , if  $f(x) = \begin{cases} \frac{e^{1/x^2}}{1 + e^{1/x^2}}, & \text{when } x \neq 0 \\ 1, & \text{when } x = 0 \end{cases}$

(Meerut 2008)

(ii) If  $f(x) = \frac{1}{x-a} \sin \frac{1}{x-a}$ , find  $f(a+0)$  and  $f(a-0)$ .

Is the function continuous at  $x = a$ ?

3. Find out the points of discontinuity of the following functions:

(i)  $f(x) = (2 + e^{1/x})^{-1} + \cos e^{1/x}$  for  $x \neq 0$ ,  $f(0) = 0$ .

(ii)  $f(x) = 1/2^n$  for  $1/2^{n-1} < x \leq 1/2^n$ ,  $n = 0, 1, 2, \dots$  and  $f(0) = 0$ .

4. If  $f(x) = \frac{1}{x} \sin \frac{1}{x}$  for  $x \neq 0$  and  $f(0) = 0$ , show that  $f(x)$  is finite for every value of  $x$  in the interval  $[-1, 1]$  but is not bounded. Determine the points of discontinuity of the function if any.

5. A function  $f$  defined on  $[0, 1]$  is given by  $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 1-x, & \text{if } x \text{ is irrational.} \end{cases}$

### LIMITS AND CONTINUITY

Show that  $f$  takes every value between 0 and 1 (both inclusive), but it is continuous only at the point  $x = \frac{1}{2}$ .

6. Prove that the function  $f$  defined by

$$f(x) = \begin{cases} \frac{1}{2}, & \text{if } x \text{ is rational} \\ \frac{1}{3}, & \text{if } x \text{ is irrational} \end{cases}$$

is discontinuous everywhere.

7. (i) Show that the function  $f$  defined by  $f(x) = \frac{x e^{1/x}}{1 + e^{1/x}}$ ,  $x \neq 0$ ,  $f(0) = 1$

is not continuous at  $x = 0$  and also show how the discontinuity can be removed.

(Meerut 2011; Rohilkhand 2006)

- (ii) Show that the function  $f(x) = 3x^2 + 2x - 1$  is continuous for  $x = 2$ .

- (iii) Show that the function  $f(x) = (1 + 2x)^{1/2}$ ,  $x \neq 0$ ,  $f(x) = e^2$ ,  $x = 0$  is continuous at  $x = 0$ .

8. Examine the continuity of the function

$$f(x) = \begin{cases} -x^2 & \text{if } x \leq 0 \\ 5x - 4 & \text{if } 0 < x \leq 1 \\ 4x^2 - 3x & \text{if } 1 < x < 2 \\ 3x + 4 & \text{if } x \geq 2 \end{cases}$$

at  $x = 0, 1$  and  $2$ .

(Meerut 2004, 06B, 07B)

9. Show that the function

$$f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1}, x \neq 0 \text{ and } f(0) = 0$$

is discontinuous at  $x = 0$ .

10. Show that the following functions are continuous at  $x = 0$ .

(i)  $f(x) = \frac{\sin x}{x}$ ,  $x \neq 0$ ,  $f(0) = 1$ .

(ii)  $f(x) = \frac{\sin^{-1} x}{x}$ ,  $x \neq 0$ ,  $f(0) = 1$ .

11. Discuss the continuity of the function  $f(x) = \frac{1}{1 + e^{1/x}}$  when  $x \neq 0$  and  $f(0) = 0$  for all values of  $x$ .

12. Prove that the function  $f(x) = \frac{|x|}{x}$  for  $x \neq 0$ ,  $f(0) = 0$  is continuous at all points except  $x = 0$ .

(Meerut 2009; Kanpur 2008)

13. Test the continuity of the function  $f(x)$  at  $x = 0$  if  $f(x) = \frac{e^{1/x} \sin(1/x)}{1 + e^{1/x}}$ ,  $x \neq 0$  and  $f(0) = 0$ .

(Meerut 2005)

14. Examine the following function for continuity at  $x = 0$  and  $x = 1$ :

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 0 \\ 1 & \text{for } 0 < x \leq 1 \\ \frac{1}{x} & \text{for } x > 1. \end{cases}$$

(Meerut 2001, 03, 04B, 05)

15. Discuss the continuity of the following function at  $x = 0$  :
- $$f(x) = \begin{cases} \cos x, & x \geq 0 \\ -\cos x, & x < 0. \end{cases}$$

(Meerut 2007)

16. Test the continuity of the following functions at  $x = 0$  :
- (i)  $f(x) = x \cos(1/x)$ , when  $x \neq 0$ ,  $f(0) = 0$ .  
 (ii)  $f(x) = x \log x$ , for  $x > 0$ ,  $f(0) = 0$ .
17. Discuss the nature of discontinuity at  $x = 0$  of the function  $f(x) = [x] - [-x]$  where  $[x]$  denotes the integral part of  $x$ .
18. Discuss the continuity of  $f(x) = (1/x) \cos(1/x)$ .
19. A function  $f$  is defined as  $f(x) = \frac{x^2 - 4x + 3}{x^2 - 1}$  for  $x \neq 1$  and  $f(x) = 2$  for  $x = 1$ .  
 Test the continuity of the function at  $x = 1$ .
20. A function  $f: \mathbb{R} \rightarrow \mathbb{R}$  is defined as  $f(x) = \frac{1}{x-4}$ .  
 Draw the graph and discuss the type of discontinuity which the function  $f(x)$  has in  $(-\infty, \infty)$ .
21. Give an example of a function  $f$  such that  $f$  is not continuous but  $|f|$  is continuous. Show that  $f^2$  can be continuous when  $f$  is not.
22. Draw a graph of the function  $y$  defined by the statement " $y$  is the smallest positive number that makes  $x + y$  an integer". For what values of  $x$  is the function discontinuous?
23. Is the function  $f$  defined by  $f(x) = (3x + 4 \tan x) / x$  continuous at  $x = 0$ ? If not, how may the function be defined to make it continuous at this point. Give reasons for your answer.
24. Give an example of each of the following types of functions :  
 (i) The function which possesses a limit at  $x = 1$  but is not defined at  $x = 1$ .  
 (ii) The function which is neither defined at  $x = 1$  nor has a limit at  $x = 1$ .  
 (iii) The function which is defined at two points but is nevertheless discontinuous at both the points.
25. In the closed interval  $[-1, 1]$  let  $f$  be defined by  
 $f(x) = x^2 \sin(1/x^2)$  for  $x \neq 0$  and  $f(0) = 0$ .  
 In the given interval (i) Is the function bounded? (ii) Is it continuous?

# ANSWERS

1. (i) Continuous for all  $x$ . (ii) Discontinuous at  $x = 0$ .  
 (iii) Discontinuous at  $x = 0$ . (iv) Continuous for all  $x$ .  
 (v) Discontinuous at  $x = 0$ . (vi) Discontinuous at 0.  
 (vii) Continuous for all  $x$ . (viii) Discontinuous at 0.  
 (ix) Discontinuous at 0. (x) Discontinuous at 0.  
 (xi) Continuous for all  $x$ .
2. (i) No, it has a discontinuity of second kind. Here both  $f(a+0)$  and  $f(a-0)$  do not exist.
3. (i) Discontinuous at  $x = 0$ .  
 (ii) Discontinuous at  $x = 1/2^n, n = 1, 2, 3, \dots$

4. Discontinuous at 0.  
 8. Continuous at  $x = 1, 2$  and discontinuous at  $x = 0$ .  
 11. Discontinuous only at  $x = 0$  and the discontinuity is ordinary.  
 13. Discontinuity of the second kind at  $x = 0$ .  
 14. Discontinuous at  $x = 0$  and continuous at  $x = 1$ .  
 15. Discontinuous at  $x = 0$ .  
 16. (i) Continuous.  
 (ii) Continuous.  
 17. Discontinuity of the first kind.  
 18. Continuous for all  $x$ , except at  $x = 0$  where it has discontinuity of the second kind.  
 19. Discontinuous at  $x = 1$ .  
 20. The function has infinite discontinuity at  $x = 4$  and is continuous at all other points of  $\mathbb{R}$ .  
 22. Discontinuous at  $x = 0, \pm 1, \pm 2, \pm 3, \dots$ .  
 23. No; define  $f(0) = 7$  to make it continuous.  
 24. (i)  $f(x) = x^2$  for  $x > 1$ ,  $f(x) = x^3$  for  $x < 1$ .  
 (ii)  $f(x) = -x^2$  for  $x < 1$ ,  $f(x) = x^2$  for  $x > 1$ .  
 (iii)  $f(x) = 0$  for  $x \leq 0$ ,  $f(x) = \frac{3}{2} - x$  for  $0 < x \leq \frac{1}{2}$ ,  $f(x) = \frac{3}{2} + x$  for  $x > \frac{1}{2}$ .  
 25. (i) Yes; (ii) Yes.

## EXERCISE 3 (Objective Questions)

### Fill In The Blanks.

Fill in the blanks "... ..", so that the following statements are complete and correct.

1. A function  $f(x)$  is continuous at a point  $x = a$  if  $\lim_{x \rightarrow a} f(x) = \dots\dots\dots$ .  
 (Bundelkhand 2008)
2.  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \dots\dots\dots$
3.  $\lim_{x \rightarrow 0} \frac{\sin(x/4)}{x} = \dots\dots\dots$
4.  $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = \dots\dots\dots$
5.  $\lim_{x \rightarrow 0} (1 + x)^{1/x} = \dots\dots\dots$
6.  $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \dots\dots\dots$
7. If  $f(x) = x - [x]$ , where  $[x]$  denotes the greatest integer less than or equal to  $x$ , then  $f(x) = \dots\dots\dots$ , for  $3 < x < 4$ .
8. Let  $f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 3 - x, & 1 \leq x \leq 2. \end{cases}$

# 2

# DIFFERENTIABILITY

## § 1. DEFINITIONS.

### Derivative at a point.

(Bundelkhand 2010)

Let  $I$  denote the open interval  $]a, b[$  in  $\mathbf{R}$  and let  $x_0 \in I$ . Then a function  $f: I \rightarrow \mathbf{R}$  is said to be **differentiable** (or **derivable**) at  $x_0$  iff

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \text{ or equivalently } \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists finitely and this limit, if it exists finitely, is called the **differential coefficient** or **derivative** of  $f$  with respect to  $x$  at  $x = x_0$ .

It is denoted by  $f'(x_0)$  or by  $Df(x_0)$ .

### Progressive and regressive derivatives.

The **progressive derivative** of  $f$  at  $x = x_0$  is given by

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}, h > 0.$$

It is also called the **right hand differential coefficient** of  $f$  at  $x = x_0$  and is denoted by  $Rf'(x_0)$  or by  $f'(x_0 + 0)$ .

The **regressive derivative** of  $f$  at  $x = x_0$  is given by

$$\lim_{h \rightarrow 0} \frac{f(x_0 - h) - f(x_0)}{-h}, h > 0.$$

It is also called the **left hand differential coefficient** of  $f$  at  $x = x_0$  and is denoted by  $Lf'(x_0)$  or by  $f'(x_0 - 0)$ .

It is obvious that  $f$  is derivable at  $x_0$  iff  $Lf'(x_0)$  and  $Rf'(x_0)$  both exist and are equal.

**Remark.** If  $f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n$  is a polynomial in  $x$  of degree  $n$ , then  $f(x)$  is differentiable at every point  $a$  of  $\mathbf{R}$ .

### Differentiability in an interval.

(Meerut 2003)

**Open interval  $]a, b[$ .** A function  $f: ]a, b[ \rightarrow \mathbf{R}$  is said to be **differentiable** in  $]a, b[$  iff it is differentiable at every point of  $]a, b[$ .

**Closed interval  $[a, b]$ .** A function  $f: [a, b] \rightarrow \mathbf{R}$  is said to be **differentiable** in  $[a, b]$  iff  $Rf'(a)$  exists,  $Lf'(b)$  exists and  $f$  is differentiable at every point of  $]a, b[$ .

**Derivative of a function.** Let  $f$  be a function whose domain is an interval  $I$ . Let  $I_1$  be the set of all those points  $x$  of  $I$  at which  $f$  is differentiable i.e.,  $f'(x)$  exists and if  $I_1 \neq \emptyset$ , we get another function  $f'$  with domain  $I_1$ . It is called the *first derivative* of  $f$  (or simply the derivative of  $f$ ). Similarly 2nd, 3rd, ...,  $n$ th derivatives of  $f$  are defined and are denoted by  $f'', f''', \dots, f^{(n)}$  respectively.

**Note.** The derivative of a function at a point and the derivative of a function are two different but related concepts. The derivative of  $f$  at a point  $a$  is a number while the derivative of  $f$  is a function. However, very often the term derivative of  $f$  is used to denote both number and function and it is left to the context to distinguish what is intended.

**An alternate definition of differentiability.**

Let  $f$  be a function defined on an interval  $I$  and let  $a$  be an interior point of  $I$ . Then, by the definition of  $f'(a)$ , assuming it to exist, we have

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

i.e.,  $f'(a)$  exists if for a given  $\varepsilon > 0$ , there exists a  $\delta > 0$  such that

$$\left| \frac{f(x) - f(a)}{x - a} - f'(a) \right| < \varepsilon \text{ whenever } 0 < |x - a| < \delta$$

or equivalently

$$x \in ]a - \delta, a + \delta[ \Rightarrow f'(a) - \varepsilon < \frac{f(x) - f(a)}{x - a} < f'(a) + \varepsilon.$$

## § 2. GEOMETRICAL MEANING OF A DERIVATIVE.

We take two neighbouring points  $P[a, f(a)]$  and  $Q[a + h, f(a + h)]$  on the curve  $y = f(x)$ .

Let the chord  $PQ$  and the tangent at  $P$  meet the  $x$ -axis in  $L$  and  $T$  respectively. Let  $\angle QPX = \alpha$  and  $\angle PTX = \psi$ . Draw  $PN$  and  $QM \perp$  to  $OX$  and  $PH \perp$  to  $QM$ .

Then  $PH = NM = OM - ON = a + h - a = h$ ,  
and  $QH = QM - MH = QM - PN = f(a + h) - f(a)$ .

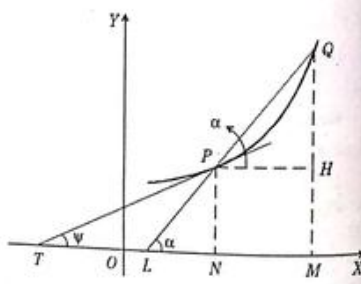
$$\therefore \tan \alpha = \frac{QH}{PH} = \frac{f(a + h) - f(a)}{h}$$

As  $h \rightarrow 0$ , the point  $Q$  moving along the curve approaches the point  $P$ , the chord  $PQ$  approaches the tangent line  $TP$  as its limiting position and the angle  $\alpha$  approaches the angle  $\psi$ .

Hence taking limits as  $h \rightarrow 0$ , the equation (1) gives

$$\tan \psi = f'(a).$$

Hence  $f'(a)$  is the tangent of the angle which the tangent line to the curve  $y = f(x)$  at the point  $P[a, f(a)]$  makes with  $x$ -axis.



## § 3. A NECESSARY CONDITION FOR THE EXISTENCE OF A FINITE DERIVATIVE.

**Theorem.** Continuity is a necessary but not a sufficient condition for the existence of a finite derivative. (Meerut 2010, 10B, 11)

**Proof.** Let  $f$  be differentiable at  $x_0$ . Then  $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$  exists and equals  $f'(x_0)$ . Now, we can write  $f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} (x - x_0)$ , if  $x \neq x_0$ .

Taking limits as  $x \rightarrow x_0$ , we get

$$\begin{aligned} \lim_{x \rightarrow x_0} [f(x) - f(x_0)] &= \lim_{x \rightarrow x_0} \left\{ \frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right\} \\ &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \lim_{x \rightarrow x_0} (x - x_0) = f'(x_0) \cdot 0 = 0. \end{aligned}$$

so that  $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ .

Hence  $f$  is continuous at  $x_0$ . Thus continuity is a necessary condition for differentiability but it is not a sufficient condition for the existence of a finite derivative. The following example illustrates this fact :

Let  $f(x) = x \sin(1/x)$ ,  $x \neq 0$  and  $f(0) = 0$ .

This function is continuous at  $x = 0$  but not differentiable at  $x = 0$ .

Since  $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 = f(0)$ , therefore the function  $f(x)$  is continuous at  $x = 0$ .

$$\begin{aligned} \text{Now } Rf'(0) &= \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h \sin(1/h) - 0}{h} = \lim_{h \rightarrow 0} \sin \frac{1}{h}, \end{aligned}$$

which does not exist. Similarly  $Lf'(0)$  does not exist.

Thus  $f(x)$  is not differentiable at  $x = 0$ , though it is continuous there.

## § 4. ALGEBRA OF DERIVATIVES.

Now we shall establish some fundamental theorems regarding the differentiability of the sum, product and quotient of differentiable functions.

**Theorem 1.** If a function  $f$  is differentiable at a point  $x_0$  and  $c$  is any real number, then the function  $cf$  is also differentiable at  $x_0$  and  $(cf)'(x_0) = cf'(x_0)$ .

**Proof.** By the definition of  $f'(x_0)$ , we have

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

$$\text{Now } \lim_{x \rightarrow x_0} \frac{(cf)(x) - (cf)(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{cf(x) - cf(x_0)}{x - x_0}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow x_0} \left\{ c \cdot \frac{f(x) - f(x_0)}{x - x_0} \right\} \\
 &= c \cdot \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = cf'(x_0).
 \end{aligned}$$

Hence  $cf$  is differentiable at  $x_0$  and  $(cf)'(x_0) = cf'(x_0)$ .

**Theorem 2.** Let  $f$  and  $g$  be defined on an interval  $I$ . If  $f$  and  $g$  are differentiable at  $x_0 \in I$ , then so also is  $f + g$  and

$$(f + g)'(x_0) = f'(x_0) + g'(x_0).$$

**Proof.** Since  $f$  and  $g$  are differentiable at  $x_0$ , therefore

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0), \quad \dots(1)$$

$$\lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0). \quad \dots(2)$$

and

$$\begin{aligned}
 \text{Now } \lim_{x \rightarrow x_0} \frac{(f + g)(x) - (f + g)(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{[f(x) + g(x)] - [f(x_0) + g(x_0)]}{x - x_0} \\
 &= \lim_{x \rightarrow x_0} \left[ \frac{f(x) - f(x_0)}{x - x_0} + \frac{g(x) - g(x_0)}{x - x_0} \right] \\
 &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} + \lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0},
 \end{aligned}$$

as the limit of a sum is equal to the sum of the limits

$$= f'(x_0) + g'(x_0), \text{ using (1) and (2).}$$

Hence  $f + g$  is differentiable at  $x_0$  and

$$(f + g)'(x_0) = f'(x_0) + g'(x_0).$$

**Theorem 3.** Let  $f$  and  $g$  be defined on an interval  $I$ . If  $f$  and  $g$  are differentiable at  $x_0 \in I$ , then so also is  $fg$  and  $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$ .

**Proof.** Since  $f$  and  $g$  are differentiable at  $x_0$ , we have

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) \quad \dots(1)$$

and

$$\lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0) \quad \dots(2)$$

Now

$$\begin{aligned}
 \lim_{x \rightarrow x_0} \frac{(fg)(x) - (fg)(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} \\
 &= \lim_{x \rightarrow x_0} \frac{f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0}
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow x_0} \left[ \frac{f(x) - f(x_0)}{x - x_0} \cdot g(x) + f(x_0) \frac{g(x) - g(x_0)}{x - x_0} \right] \\
 &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \lim_{x \rightarrow x_0} g(x) + f(x_0) \cdot \lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0} \\
 &= f'(x_0)g(x_0) + f(x_0)g'(x_0),
 \end{aligned}$$

using (1), (2) and the fact that  $\lim_{x \rightarrow x_0} g(x) = g(x_0)$ .

Note that  $g(x)$  is differentiable at  $x = x_0$  implies that  $g(x)$  is continuous at  $x_0$  and so

$$\lim_{x \rightarrow x_0} g(x) = g(x_0).$$

Hence  $fg$  is differentiable at  $x_0$  and

$$(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

**Theorem 4.** If  $f$  is differentiable at  $x_0$  and  $f(x_0) \neq 0$ , then the function  $1/f$  is differentiable at  $x_0$  and  $(1/f)'(x_0) = -f'(x_0)/[f(x_0)]^2$ .

**Proof.** Since  $f$  is differentiable at  $x_0$ , therefore  $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$ .

...

Since  $f$  is differentiable at  $x_0$ , it is continuous at  $x_0$ , therefore

$$\lim_{x \rightarrow x_0} f(x) = f(x_0) \neq 0. \quad \dots(2)$$

Also, since  $f(x_0) \neq 0$ , hence,  $f(x_0) \neq 0$  in some neighbourhood  $N$  of  $x_0$ . Now, we have for  $x \in N$

$$\begin{aligned}
 \lim_{x \rightarrow x_0} \frac{1}{f(x)} - \frac{1}{f(x_0)} &= \lim_{x \rightarrow x_0} \left\{ \frac{f(x) - f(x_0)}{x - x_0} \cdot \frac{1}{f(x)} \cdot \frac{1}{f(x_0)} \right\} \\
 &= - \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \lim_{x \rightarrow x_0} \frac{1}{f(x)} \cdot \frac{1}{f(x_0)} \\
 &= -f'(x_0) \cdot \frac{1}{f(x_0)} \cdot \frac{1}{f(x_0)}, \text{ using (1) and (2)} \\
 &= -f'(x_0)/[f(x_0)]^2.
 \end{aligned}$$

Hence  $1/f$  is differentiable at  $x_0$  and

$$(1/f)'(x_0) = -f'(x_0)/[f(x_0)]^2.$$

**Theorem 5.** Let  $f$  and  $g$  be defined on an interval  $I$ . If  $f$  and  $g$  are differentiable at  $x_0 \in I$ , and  $g(x_0) \neq 0$ , then the function  $f/g$  is differentiable at  $x_0$  and

$$\left( \frac{f}{g} \right)'(x_0) = \frac{[g(x_0)f'(x_0) - f(x_0)g'(x_0)]}{[g(x_0)]^2}.$$

**Proof.** Use theorems 3 and 4 of § 5.

$= \lim_{h \rightarrow 0} \left( 2h \sin \frac{1}{h} - \cos \frac{1}{h} \right)$ , which does not exist.

Similarly it can be shown that  $f'(0-0)$  does not exist. Hence  $f'$  is discontinuous at the origin. Since both the limits  $f'(0-0)$  and  $f'(0+0)$  do not exist, therefore the discontinuity is of the second kind.

**Ex. 11.** A function  $f$  is defined by  $f(x) = x^p \cos(1/x)$ ,  $x \neq 0$ ;  $f(0) = 0$ . What conditions should be imposed on  $p$  so that  $f$  may be

- (i) continuous at  $x = 0$   
(ii) differentiable at  $x = 0$ ?

**Sol.** We have

$$f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} [(0+h)^p \cos \{1/(0+h)\}]$$

$$= \lim_{h \rightarrow 0} h^p \cos(1/h)$$

and  $f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} [(0-h)^p \cos \{1/(0-h)\}]$

$$= \lim_{h \rightarrow 0} (-h)^p \cos(1/h).$$

Now if the function  $f(x)$  is to be continuous at  $x = 0$ , then

$$f(0+0) = f(0) = 0 = f(0-0)$$

i.e., the limits given in (1) and (2) must both tend to zero.

This is possible only if  $p > 0$ , which is the required condition.

Now  $Rf'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$

$$= \lim_{h \rightarrow 0} \frac{h^p \cos(1/h) - 0}{h} = \lim_{h \rightarrow 0} h^{p-1} \cos \frac{1}{h}$$

and  $Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{(-h)^p \cos(-1/h) - 0}{-h}$

$$= \lim_{h \rightarrow 0} -(-1)^p h^{p-1} \cos(1/h).$$

Now if  $f'(x)$  exists at  $x = 0$  then we must have  $Rf'(0) = Lf'(0)$  and this is possible only if  $p-1 > 0$  i.e.,  $p > 1$  which gives  $Rf'(0) = 0 = Lf'(0)$ . Hence in order that  $f$  is differentiable at  $x = 0$ ,  $p$  must be greater than 1.

**Ex. 12.** For a real number  $y$ , let  $[y]$  denote the greatest integer less than or equal to  $y$ . Then if  $f(x) = \frac{\tan(\pi[x-\pi])}{1+[x]^2}$ , show that  $f'(x)$  exists for all  $x$ .

**Sol.** From the definition of  $[y]$ , we see that  $[x-\pi]$  is an integer for all values of  $x$ . Then  $\pi(x-\pi)$  is an integral multiple of  $\pi$  and so  $\tan(\pi[x-\pi]) = 0 \forall x$ . Since  $[x]$  is an integer so  $1+[x]^2 \neq 0$  for any  $x$ . Thus  $f(x) = 0$  for all  $x$  i.e.,  $f(x)$  is a constant function and so it is continuous and differentiable i.e.,  $f'(x)$  exists  $\forall x$  and is equal to zero.

**Ex. 13.** Determine the set of all points where the function  $f(x) = x/(1+[x])$  is differentiable.

**Sol.** Since  $|x| = x, x > 0$ ,  $|x| = -x, x < 0$ ,  $|x| = 0, x = 0$ ,

$$\therefore f(x) = \frac{x}{1-x}, x < 0; f(x) = 0, x = 0; f(x) = \frac{x}{1+x}, x > 0.$$

We have  $f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \frac{h}{1+h} = 0$ ;

$$f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} \frac{-h}{1-h} = 0.$$

Since  $f(0+0) = f(0) = f(0-0) = 0$  so the function is continuous at  $x = 0$ .

Further  $Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{[-h/(1-h)] - 0}{-h} = \lim_{h \rightarrow 0} \frac{1}{1-h} = 1;$$

$$Rf'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{[h/(1+h)] - 0}{h} = 1.$$

Since  $Lf'(0) = Rf'(0)$ , so the function is differentiable at  $x = 0$ . It is obviously differentiable for all other real values of  $x$ . Hence it is differentiable in the interval  $]-\infty, \infty[$ .

**Ex. 14.** Let  $f(x) = \sqrt{x} \{1 + x \sin(1/x)\}$  for  $x > 0$ ,  $f(0) = 0$ .

$$f(x) = -\sqrt{-x} \{1 + x \sin(1/x)\} \text{ for } x < 0.$$

Show that  $f'(x)$  exists everywhere and is finite except at  $x = 0$  where its value is  $+\infty$ .

**Sol.** We have

$$Rf'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{h}) \{1 + h \sin(1/h)\} - 0}{h}$$

$$= \lim_{h \rightarrow 0} \left[ \frac{1}{\sqrt{h}} + (\sqrt{h}) \sin(1/h) \right] = \infty + 0 = \infty$$

and  $Lf'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{-\sqrt{-h} \{1 + (-h) \sin(1/(-h))\} - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \left[ \frac{1}{\sqrt{-h}} + (\sqrt{-h}) \sin \frac{1}{h} \right] = \infty + 0 = \infty$$

Since  $Rf'(0) = Lf'(0) = \infty$ ,  $\therefore f'(0) = \infty$ .

We have  $f'(x) = \frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} \sin \frac{1}{x} - \frac{1}{\sqrt{x}} \cos \frac{1}{x}$  for  $x > 0$

and  $f'(x) = \frac{1}{2\sqrt{-x}} + \frac{3}{2}\sqrt{-x} \sin \frac{1}{x} - \frac{1}{\sqrt{-x}} \cos \frac{1}{x}$  for  $x < 0$ .

Hence  $f'(a)$  is finite for all  $a \neq 0$ .

**Ex. 15.** Draw the graph of the function  $y = |x-1| + |x-2|$  in the interval  $[0, 3]$  and discuss the continuity and differentiability of the function in this interval. (Meerut 2007B, 09)

**Sol.** From the given definition of the function, we have

$$\begin{aligned}
 y &= 1 - x - 2 = 3 - 2x & \text{when } x \leq 1 \\
 y &= x - 1 - 2 = x - 3 & \text{when } 1 \leq x \leq 2 \\
 y &= x - 1 + x - 2 = 2x - 3 & \text{when } x \geq 2.
 \end{aligned}$$

Thus the graph consists of the segments of the three straight lines  $y = 3 - 2x$ ,  $y = x - 3$  and  $y = 2x - 3$  respectively. The graph of the function for the interval  $[0, 3]$  is as given in the figure.

The graph shows that the function is continuous throughout the interval but is not differentiable at  $x = 1, 2$  because the slopes at these points are different on the left and right hand sides.

To test it analytically, we write  $y = f(x)$ . Then

$$\begin{aligned}
 f(x) &= 3 - 2x & \text{when } x \leq 1 \\
 &= x - 3 & \text{when } 1 \leq x \leq 2 \\
 &= 2x - 3 & \text{when } x \geq 2.
 \end{aligned}$$

This function is obviously continuous and differentiable at all points of the interval  $[0, 3]$  except possibly at  $x = 1$  and at  $x = 2$ .

At  $x = 1$ , we have  $f(1) = 1$ :

$$f(1 - 0) = \lim_{h \rightarrow 0} [3 - 2(1 - h)] = 1; f(1 + 0) = \lim_{h \rightarrow 0} (1) = 1.$$

Since  $f(1 - 0) = f(1 + 0) = f(1)$ ,  $f$  is continuous at  $x = 1$ .

$$\text{Again } Rf'(1) = \lim_{h \rightarrow 0} \frac{f(1 + h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{1 - 1}{h} = 0$$

$$\text{and } Lf'(1) = \lim_{h \rightarrow 0} \frac{f(1 - h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{3 - 2(1 - h) - 1}{-h} = -2.$$

Since  $Rf'(1) \neq Lf'(1)$ ,  $f$  is not differentiable at  $x = 1$ .

At  $x = 2$ , we have  $f(2) = 1$ :

$$f(2 - 0) = \lim_{h \rightarrow 0} (1) = 1; f(2 + 0) = \lim_{h \rightarrow 0} [2(2 + h) - 3] = 1.$$

Since  $f(2 - 0) = f(2 + 0) = f(2)$ ,  $f$  is continuous at  $x = 2$ .

$$\text{Again } Rf'(2) = \lim_{h \rightarrow 0} \frac{f(2 + h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{2(2 + h) - 3 - 1}{h} = 2$$

$$\text{and } Lf'(2) = \lim_{h \rightarrow 0} \frac{f(2 - h) - f(2)}{-h} = \lim_{h \rightarrow 0} \frac{1 - 1}{-h} = 0.$$

Since  $Rf'(2) \neq Lf'(2)$ ,  $f$  is not differentiable at  $x = 2$ .

**Ex. 16.** Show that the function  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by

$$f(x) = x \left[ 1 + \frac{1}{2} \sin \log x^2 \right], x \neq 0 \text{ and } f(0) = 0$$

is everywhere continuous but has no differential coefficient at the origin.

**Sol.** Obviously the function  $f(x)$  is continuous at every point of  $\mathbb{R}$  except possibly at  $x = 0$ . We test at  $x = 0$ . Given  $f(0) = 0$ ,

$$f(0 + 0) = \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} \left[ (0 + h) \left\{ 1 + \frac{1}{2} \sin \log (0 + h)^2 \right\} \right]$$

$$= \lim_{h \rightarrow 0} [h + (h/2) \sin \log h^2] = 0 + 0 \times \text{a finite quantity} = 0.$$

[ $\because \sin \log h^2$  oscillates between  $-1$  and  $+1$  as  $h \rightarrow 0$ ]

Similarly we can show that  $f(0 - 0) = 0$ .

Hence  $f$  is continuous at  $x = 0$ .

$$\text{Now } Rf'(0) = \lim_{h \rightarrow 0} \frac{(0 + h) \left\{ 1 + \frac{1}{2} \sin \log (0 + h)^2 \right\} - 0}{h}$$

$$= \lim_{h \rightarrow 0} \left\{ 1 + \frac{1}{2} \sin \log h^2 \right\}, \text{ which does not exist since } \sin \log h^2$$

oscillates between  $-1$  and  $+1$  as  $h \rightarrow 0$ .

$$Lf'(0) = \lim_{h \rightarrow 0} \frac{(0 - h) \left\{ 1 + \frac{1}{2} \sin \log (0 - h)^2 \right\} - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \left\{ 1 + \frac{1}{2} \sin \log h^2 \right\}, \text{ which does not exist as above.}$$

Hence  $f$  has no differential coefficient at  $x = 0$ .

**Ex. 17.** Let  $f(x) = x \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}$ ,  $x \neq 0$ ;  $f(0) = 0$ .

Show that  $f(x)$  is continuous but not derivable at  $x = 0$ .

(Meerut 2005)

**Sol.** We have  $f(0) = 0$ :

$$\begin{aligned}
 f(0 + 0) &= \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} h \frac{e^{1/h} - e^{-1/h}}{e^{1/h} + e^{-1/h}} \\
 &= \lim_{h \rightarrow 0} h \frac{1 - e^{-2/h}}{1 + e^{-2/h}}, \text{ dividing the Nr. and Dr. by } e^{1/h} \\
 &= 0 \times \frac{1 - 0}{1 + 0} = 0 \times 1 = 0;
 \end{aligned}$$

and

$$\begin{aligned}
 f(0 - 0) &= \lim_{h \rightarrow 0} f(0 - h) = \lim_{h \rightarrow 0} f(-h) \\
 &= \lim_{h \rightarrow 0} -h \frac{e^{1/-h} - e^{-1/-h}}{e^{1/-h} + e^{-1/-h}} = \lim_{h \rightarrow 0} -h \frac{e^{-1/h} - e^{1/h}}{e^{-1/h} + e^{1/h}} \\
 &= \lim_{h \rightarrow 0} -h \frac{e^{-2/h} - 1}{e^{-2/h} + 1} = 0 \times \frac{0 - 1}{0 + 1} = 0.
 \end{aligned}$$

Since  $f(0 + 0) = f(0 - 0) = f(0)$ , the function is continuous at  $x = 0$ .

$$\begin{aligned}
 \text{Now } Rf'(0) &= \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\
 &= \lim_{h \rightarrow 0} \left[ \frac{h \frac{e^{1/h} - e^{-1/h}}{e^{1/h} + e^{-1/h}} - 0}{h} \right] = \lim_{h \rightarrow 0} \frac{1 - e^{-2/h}}{1 + e^{-2/h}} = \frac{1 - 0}{1 + 0} = 1.
 \end{aligned}$$

and

$$\begin{aligned}
 Lf'(0) &= \lim_{h \rightarrow 0} \frac{f(0 - h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h} \\
 &= \lim_{h \rightarrow 0} \left[ \frac{(-h) \frac{e^{-1/h} - e^{1/h}}{e^{-1/h} + e^{1/h}} - 0}{-h} \right] = \lim_{h \rightarrow 0} \left[ \frac{e^{-1/h} - e^{1/h}}{e^{-1/h} + e^{1/h}} \right]
 \end{aligned}$$

$$\lim_{h \rightarrow 0} \frac{e^{1/h} - 1}{e^{1/h} + 1} = \frac{0 - 1}{0 + 1} = -1$$

Since  $f'(0) = 1/f'(0)$ , the function is not derivable at  $x = 0$ .

Ex. 18. Let  $f(x) = e^{-1/x^2} \sin(1/x)$  when  $x \neq 0$  and  $f(0) = 0$ . Show that at  $x = 0$ ,  $f$  has a differential coefficient and this is continuous at  $x = 0$ .

Sol. We use differentiability at  $x = 0$ .

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{e^{-1/h^2} \sin(1/h) - 0}{h} = \lim_{h \rightarrow 0} \frac{\sin(1/h)}{h e^{1/h^2}} \\ &= \lim_{h \rightarrow 0} \frac{\sin(1/h)}{h \left[ 1 + \frac{1}{h^2} + \frac{1}{2! h^4} + \dots \right]} = \lim_{h \rightarrow 0} \frac{\sin(1/h)}{h + \frac{1}{h} + \frac{1}{2! h^3} + \dots} \\ &= \text{a finite quantity between } -1 \text{ and } 1 = 0. \end{aligned}$$

Similarly  $f''(0) = 0$ .

Since  $f'(0) = 1/f'(0) = 0$ , hence the function  $f(x)$  is differentiable at  $x = 0$  and  $f'(0) = 0$ .

If  $x$  is any point other than zero, then

$$\begin{aligned} f'(x) &= (2/x^3) e^{-1/x^2} \sin(1/x) - (1/x^2) e^{-1/x^2} \cos(1/x) \\ &= (2/x) \sin(1/x) - \cos(1/x) (1/x^2) (1/e^{1/x^2}) \end{aligned}$$

$$\begin{aligned} \text{Now } f'(0) &= \lim_{h \rightarrow 0} \frac{f'(0) - f'(h)}{h} = \lim_{h \rightarrow 0} \left( \frac{2}{h} \sin \frac{1}{h} - \cos \frac{1}{h} \right) \cdot \frac{1}{h^2 e^{1/h^2}} \\ &= \lim_{h \rightarrow 0} \left( \frac{2 \sin(1/h)}{h^2 e^{1/h^2}} - \frac{\cos(1/h)}{h^2 e^{1/h^2}} \right) \\ &= \lim_{h \rightarrow 0} \left[ \frac{2 \sin(1/h)}{h^2 \left( 1 + \frac{1}{h^2} + \frac{1}{2! h^4} + \dots \right)} - \frac{\cos(1/h)}{h^2 \left( 1 + \frac{1}{h^2} + \frac{1}{2! h^4} + \dots \right)} \right] \\ &= \frac{\text{some finite quantity}}{\infty} - \frac{\text{some finite quantity}}{\infty} = 0. \end{aligned}$$

Similarly  $f''(0) = 0$ . Hence  $f''$  is continuous at  $x = 0$ .

### EXERCISE 1

- Show that the function  $f(x) = |x - 1|$  is not differentiable at  $x = 1$ .  
(a) If  $f(x) = x/(1 + e^{1/x})$ ,  $x \neq 0$ ,  $f(0) = 0$ , show that  $f$  is continuous at  $x = 0$ , but  $f'(0)$  does not exist.  
(b) If  $f(x) = \frac{x e^{1/x}}{1 + e^{1/x}}$  for  $x \neq 0$  and  $f(0) = 0$ , show that  $f(x)$  is continuous at  $x = 0$ , but  $f'(0)$  does not exist.
- A function  $\phi$  is defined as follows:  
 $\phi(x) = -x$  for  $x \leq 0$ ,  $\phi(x) = x$  for  $x \geq 0$ .

### DIFFERENTIABILITY

Test the character of the function at  $x = 0$  as regards continuity and differentiability.

- Show that the function  $f$  defined on  $\mathbb{R}$  by  
 $f(x) = |x - 1| + 2|x - 2| + 3|x - 3|$   
is continuous but not differentiable at the points 1, 2 and 3.

(Bundelkhand 2009)

- Show that the function

$$\begin{aligned} f(x) &= x, & 0 < x \leq 1 \\ &= x - 1, & 1 < x \leq 2 \end{aligned}$$

has no derivative at  $x = 1$ .

- Show that the function

$$f(x) = x^2 - 1, x \geq 1 = 1 - x, x < 1$$

has no derivative at  $x = 1$ .

- The following limits are derivatives of certain functions at a certain point. Determine these functions and the points.

$$(i) \lim_{x \rightarrow 2} \frac{\log x - \log 2}{x - 2}, \quad (ii) \lim_{h \rightarrow 0} \frac{\sqrt[3]{a+h} - \sqrt[3]{a}}{h}$$

- Let  $f(x) = x^2 \sin(x^{-4/3})$  except when  $x = 0$  and  $f(0) = 0$ . Prove that  $f(x)$  has zero as a derivative at  $x = 0$ .

- A function  $\phi(x)$  is defined as follows:

$$\begin{aligned} \phi(x) &= 1 + x & \text{if } x \leq 2 \\ &= 5 - x & \text{if } x > 2. \end{aligned}$$

Test the character of the function at  $x = 2$  as regards its continuity and differentiability.

- Examine the following curve for continuity and differentiability at  $x = 0$  and  $x = 1$ :

$$\begin{aligned} y &= x^2 & \text{for } x \leq 0 \\ y &= 1 & \text{for } 0 < x \leq 1 \\ y &= 1/x & \text{for } x > 1. \end{aligned}$$

Also draw the graph of the function.

(Meerut 2004B, 09B)

- A function  $f(x)$  is defined as follows:

$$\begin{aligned} f(x) &= 1 + x & \text{for } x \leq 0, \\ f(x) &= x & \text{for } 0 < x < 1, \\ f(x) &= 2 - x & \text{for } 1 \leq x \leq 2, \\ f(x) &= 3x - x^2 & \text{for } x > 2. \end{aligned}$$

Discuss the continuity of  $f(x)$  and the existence of  $f'(x)$  at  $x = 0, 1$  and  $2$ .

- Discuss the continuity and differentiability of the following function:

$$\begin{aligned} f(x) &= x^2 & \text{for } x < -2 \\ f(x) &= 4 & \text{for } -2 \leq x \leq 2 \\ f(x) &= x^2 & \text{for } x > 2. \end{aligned}$$

Also draw the graph.

(Meerut 2007, 10B)

13. A function  $f(x)$  is defined as follows :

$$f(x) = x \quad \text{for } 0 \leq x \leq 1$$

$$f(x) = 2 - x \quad \text{for } x \geq 1.$$

Test the character of the function at  $x = 1$  as regards the continuity and differentiability. (Meerut 2003)

14. Examine the function defined by
- $$f(x) = x^2 \cos(e^{1/x}), x \neq 0,$$
- $$f(0) = 0$$

with regard to (i) continuity (ii) differentiability in the interval  $[-1, 1]$ .

15. (a) Define continuity and differentiability of a function at a given point. If a function possesses a finite differential coefficient at a point, show that it is continuous at this point. Is the converse true? Give example in support of your answer.
- (b) What do you understand by the derivative of a real valued function at a point? Show that  $f(x) = x \sin(1/x), x \neq 0, f(0) = 0$  is not derivable at  $x = 0$ .
- (c) Prove that if a function  $f(x)$  possesses a finite derivative in a closed interval  $[a, b]$ , then  $f(x)$  is continuous in  $[a, b]$ .

### ANSWERS

3. Continuous at  $x = 0$  but not differentiable at  $x = 0$ .
7. (i) The function is  $\log x$  and the point is  $x = 2$ .
- (ii) The function is  $\sqrt{x}$  and the point is  $x = a$ .
9. Continuous but not differentiable at  $x = 2$ .
10. Discontinuous and non-differentiable at  $x = 0$ , continuous and non-differentiable at  $x = 1$ .
11. Discontinuous and non-differentiable at  $x = 0, 2$  and continuous but not differentiable at  $x = 1$ .
12. Continuous but not differentiable at  $x = -2, 2$ .
13. Continuous but non-differentiable at  $x = 1$ .
14. Continuous and differentiable throughout  $\mathbb{R}$ .

### § 5. ROLLE'S THEOREM.

If a function  $f(x)$  is such that

- (i)  $f(x)$  is continuous in the closed interval  $[a, b]$ ,
- (ii)  $f'(x)$  exists for every point in the open interval  $]a, b[$ ,
- (iii)  $f(a) = f(b)$ , then there exists at least one value of  $x$ , say  $c$ , where  $a < c < b$ , such that  $f'(c) = 0$ .

(Kanpur 2008)

**Proof.** Since  $f$  is continuous on  $[a, b]$ , it is bounded on  $[a, b]$ . Let  $M$  and  $m$  be the supremum and infimum of  $f$  respectively in the closed interval  $[a, b]$ .

Now either  $M = m$  or  $M \neq m$ .

### DIFFERENTIABILITY

If  $M = m$ , then  $f$  is a constant function over  $[a, b]$  and consequently  $f'(x) = 0$  for all  $x$  in  $[a, b]$ . Hence the theorem is proved in this case.

If  $M \neq m$ , then at least one of the numbers  $M$  and  $m$  must be different from the equal values  $f(a)$  and  $f(b)$ . For the sake of definiteness, let  $M \neq f(a)$ .

Since every continuous function on a closed interval attains its supremum, therefore, there exists a real number  $c$  in  $[a, b]$  such that  $f(c) = M$ . Also, since  $f(a) \neq M \neq f(b)$ , therefore,  $c$  is different from both  $a$  and  $b$ . This implies that  $c \in ]a, b[$ .

Now  $f(c)$  is the supremum of  $f$  on  $[a, b]$ , therefore,

$$f(x) \leq f(c) \quad \forall x \text{ in } [a, b].$$

In particular, for all positive real numbers  $h$  such that  $c - h$  lies in  $[a, b]$ ,

$$f(c - h) \leq f(c)$$

$$\frac{f(c - h) - f(c)}{-h} \geq 0.$$

$\Rightarrow$  ... (2)

Since  $f'(x)$  exists at each point of  $]a, b[$ , and hence, in particular  $f'(c)$  exists, so taking limit as  $h \rightarrow 0$ , (2) gives  $Lf'(c) \geq 0$ .

Similarly, from (1), for all positive real numbers  $h$  such that  $c + h$  lies in  $[a, b]$ , we have

$$f(c + h) \leq f(c).$$

By the same argument as above, we get

$$Rf'(c) \leq 0.$$

Since  $f'(c)$  exists, hence,  $Lf'(c) = f'(c) = Rf'(c)$ .

From (3), (4) and (5) we conclude that  $f'(c) = 0$ .

In the same manner we can consider the case  $M = f(a) \neq m$ .

**Note 1.** There may be more than one point like  $c$  at which  $f'(x)$  vanishes.

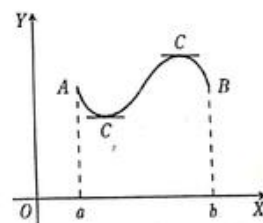
**Note 2.** Rolle's theorem will not hold good

- (i) if  $f(x)$  is discontinuous at some point in the interval  $a \leq x \leq b$ ,
- or (ii) if  $f'(x)$  does not exist at some point in the interval  $a < x < b$ ,
- or (iii) if  $f(a) \neq f(b)$ .

**Note 3.** It can be seen that the conditions of Rolle's theorem are not necessary for  $f'(x)$  to vanish at some point in  $]a, b[$ . For example,  $f(x) = \cos(1/x)$  is discontinuous at  $x = 0$  in the interval  $[-1, 2]$  but  $f'(x)$  vanishes at an infinite number of points in the interval.

### Geometrical interpretation of Rolle's Theorem.

Suppose the function  $f(x)$  is not constant and satisfies the conditions of Rolle's theorem in the interval  $[a, b]$ . Then its geometrical interpretation is that the curve representing the graph of the function  $f$  must have a tangent parallel to  $x$ -axis, at least at one point between  $a$  and  $b$ .



### Algebraical interpretation of Rolle's Theorem.

Rolle's theorem leads to a very important result in the theory of equations. When  $f(a) = f(b) = 0$  and  $f: [a, b] \rightarrow \mathbb{R}$  is a polynomial function  $f(x)$ . Here  $a$  and  $b$  are the roots of the equation  $f(x) = 0$ . Since a polynomial function  $f(x)$  is continuous and differentiable at every point of its domain and we have taken  $f(a) = f(b) = 0$ , therefore, all the three conditions of Rolle's theorem are satisfied and consequently there exists a point  $c \in [a, b]$  such that  $f'(c) = 0$  i.e., if  $a$  and  $b$  are any two roots of the polynomial equation  $f(x) = 0$ , then there exists at least one root of the equation  $f'(x) = 0$  which lies between  $a$  and  $b$ .

### SOLVED EXAMPLES

**Ex. 1.** Discuss the applicability of Rolle's theorem for  $f(x) = 2 + (x-1)^{2/3}$  in the interval  $[0, 2]$ .

**Sol.** We have  $f(x) = 2 + (x-1)^{2/3}$ . Here  $f(0) = 3 = f(2)$ , which shows that the third condition of Rolle's theorem is satisfied.

Since  $f(x)$  is an algebraic function of  $x$ , it is continuous in the closed interval  $[0, 2]$ . Thus the first condition of Rolle's theorem is satisfied.

Now  $f'(x) = \frac{2}{3} [1/(x-1)^{1/3}]$ . We see that for  $x = 1$ ,  $f'(x)$  is not finite while  $x = 1$  is a point of the open interval  $0 < x < 2$ . Thus the second condition of Rolle's theorem is not satisfied.

Hence the Rolle's theorem is not applicable for the function  $f(x) = 2 + (x-1)^{2/3}$  in the interval  $[0, 2]$ .

**Ex. 2.** Discuss the applicability of Rolle's theorem in the interval  $[-1, 1]$  to the function  $f(x) = |x|$ .

**Sol.** Given  $f(x) = |x|$ . Here  $f(-1) = |-1| = 1$ ,  $f(1) = |1| = 1$ , so that  $f(-1) = f(1)$ .

Further the function  $f(x)$  is continuous throughout the closed interval  $[-1, 1]$  but it is not differentiable at  $x = 0$  which is a point of the open interval  $]-1, 1[$ . Thus the second condition of Rolle's theorem is not satisfied. Hence the Rolle's theorem is not applicable here.

**Ex. 3.** Are the conditions of Rolle's theorem satisfied in the case of the following functions?

(i)  $f(x) = x^2$  in  $2 \leq x \leq 3$ .

(ii)  $f(x) = \cos(1/x)$  in  $-1 \leq x \leq 1$ .

(iii)  $f(x) = \tan x$  in  $0 \leq x \leq \pi$ .

**Sol.** (i) The function  $f(x) = x^2$  is continuous and differentiable in the interval  $[2, 3]$ . Also  $f(2) = 4$  and  $f(3) = 9$ , so that  $f(2) \neq f(3)$ .

Thus the first two conditions of Rolle's theorem are satisfied and the third condition is not satisfied.

(ii) The function  $f(x) = \cos(1/x)$  is discontinuous at  $x = 0$  and consequently is not differentiable there. Thus the first two conditions of Rolle's theorem are not satisfied.

### DIFFERENTIABILITY

Here  $f(-1) = \cos(-1) = \cos 1$  and  $f(1) = \cos 1$ . Thus  $f(-1) = f(1)$  i.e., the third condition is satisfied.

(iii) The function  $f(x) = \tan x$  is not continuous at  $x = \pi/2$  and consequently is not differentiable there. Thus the first two conditions of Rolle's theorem are not satisfied here.

Further  $f(0) = \tan 0 = 0$  and  $f(\pi) = \tan \pi = 0$ . Thus  $f(0) = f(\pi)$  i.e., the third condition is satisfied.

**Ex. 4.** Discuss the applicability of Rolle's theorem to  $f(x) = \log \left[ \frac{x^2 + ab}{(a+b)x} \right]$  in the interval  $[a, b]$ ,  $0 < a < b$ .

**Sol.** Here  $f(a) = \log \left[ \frac{a^2 + ab}{(a+b)a} \right] = \log 1 = 0$ ,

and  $f(b) = \log \left[ \frac{b^2 + ab}{(a+b)b} \right] = \log 1 = 0$ .

Thus  $f(a) = f(b) = 0$ .

Also  $Rf'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \log \left[ \frac{(x+h)^2 + ab}{(a+b)(x+h)} \right] - \log \left[ \frac{x^2 + ab}{(a+b)x} \right] \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \log \left[ \frac{(x^2 + 2xh + h^2 + ab)(a+b)x}{(a+b)(x+h)(x^2 + ab)} \right] \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \log \left[ \frac{(x^2 + 2xh + h^2 + ab)}{x^2 + ab} \times \frac{x}{x+h} \right] \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \log \left[ 1 + \frac{2xh + h^2}{x^2 + ab} \right] - \log \left[ 1 + \frac{h}{x} \right] \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{2xh + h^2}{x^2 + ab} - \frac{h}{x} + \dots \right] \quad \dots (1)$$

$$\left[ \because \log(1+y) = y - \frac{1}{2}y^2 + \dots \right]$$

$$= \frac{2x}{x^2 + ab} - \frac{1}{x}$$

Again  $Lf'(x) = \lim_{h \rightarrow 0} \left[ \frac{f(x-h) - f(x)}{-h} \right]$

$$= \lim_{h \rightarrow 0} \frac{1}{(-h)} \left[ -\frac{2hx + h^2}{x^2 + ab} - \frac{(-h)}{x} + \dots \right], \text{ replacing } h \text{ by } -h \text{ in (1)}$$

$$= \frac{2x}{x^2 + ab} - \frac{1}{x}$$

Since  $Rf'(x) = Lf'(x)$ ,  $f(x)$  is differentiable for all values of  $x$  in  $[a, b]$ . This implies that  $f(x)$  is also continuous for all values of  $x$  in  $[a, b]$ . Thus all the three conditions of Rolle's theorem are satisfied. Hence  $f'(x) = 0$  for at least one value of  $x$  in the open interval  $]a, b[$ .

Now  $f'(x) = 0 \Rightarrow \frac{2x}{x^2 + ab} - \frac{1}{x} = 0$  or  $2x^2 - (x^2 + ab) = 0$   
 $x^2 = ab$  or  $x = \sqrt{ab}$ ,  
 which being the geometric mean of  $a$  and  $b$  lies in the open interval  $[a, b]$ . Hence the Rolle's theorem is verified.

**Remark.** In this question to find  $f'(x)$ , we can also proceed as follows:

We have  $f(x) = \log(x^2 + ab) - \log(a + b) - \log x$ .  
 $\therefore f'(x) = \frac{2x}{x^2 + ab} - \frac{1}{x}$

Obviously  $f'(x)$  exists for all values of  $x$  in  $[a, b]$ .

**Ex. 5.** Verify Rolle's theorem in the case of the functions

- (i)  $f(x) = 2x^3 + x^2 - 4x - 2$ ,
- (ii)  $f(x) = \sin x$  in  $[0, \pi]$ ,
- (iii)  $f(x) = (x-a)^m(x-b)^n$ , where  $m$  and  $n$  are +ive integers, and  $x$  lies in the interval  $[a, b]$ .

**Sol.** (i) Since  $f(x)$  is a rational integral function of  $x$ , therefore, it is continuous and differentiable for all real values of  $x$ . Thus the first two conditions of Rolle's theorem are satisfied in any interval.

Here  $f(x) = 0$  gives  $2x^3 + x^2 - 4x - 2 = 0$   
 $(x^2 - 2)(2x + 1) = 0$  i.e.,  $x = \pm \sqrt{2}, -\frac{1}{2}$

Thus  $f(\sqrt{2}) = f(-\sqrt{2}) = f(-\frac{1}{2}) = 0$ .

If we take the interval  $[-\sqrt{2}, \sqrt{2}]$ , then all the three conditions of Rolle's theorem are satisfied in this interval. Consequently there is at least one value of  $x$  in the open interval  $]-\sqrt{2}, \sqrt{2}[$  for which  $f'(x) = 0$ .

Now  $f'(x) = 0 \Rightarrow 6x^2 + 2x - 4 = 0 \Rightarrow 3x^2 + x - 2 = 0$   
 $(3x - 2)(x + 1) = 0$  or  $x = -1, 2/3$  i.e.,  $f'(-1) = f'(2/3) = 0$ .

Since both the points  $x = -1$  and  $x = 2/3$  lie in the open interval  $]-\sqrt{2}, \sqrt{2}[$ , Rolle's theorem is verified.

(ii) The function  $f(x) = \sin x$  is continuous and differentiable in  $[0, \pi]$ .

Also  $f(0) = 0 = f(\pi)$ . Thus all the three conditions of Rolle's theorem are satisfied. Hence  $f'(x) = 0$  for at least one value of  $x$  in the open interval  $]0, \pi[$ .

Now  $f'(x) = 0 \Rightarrow \cos x = 0 \Rightarrow x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$

Since  $x = \pi/2$  lies in the open interval  $]0, \pi[$ , the Rolle's theorem is verified.

(iii) We have  $f(x) = (x-a)^m(x-b)^n$ .

As  $m$  and  $n$  are positive integers,  $(x-a)^m$  and  $(x-b)^n$  are polynomials in  $x$  on being expanded by binomial theorem. Hence  $f(x)$  is also a polynomial in  $x$ . Consequently  $f(x)$  is continuous and differentiable in the closed interval  $[a, b]$ . Also  $f(a) = f(b) = 0$ .

Thus all the three conditions of Rolle's theorem are satisfied so that there is at least one value of  $x$  in the open interval  $]a, b[$  where  $f'(x) = 0$ .

Now  $f'(x) = (x-a)^m \cdot n(x-b)^{n-1} + m(x-a)^{m-1}(x-b)^n$ .

Solving the equation  $f'(x) = 0$ , we get  $x = a, b, (na + mb)/(m + n)$ .

Out of these values the value  $(na + mb)/(m + n)$  is a point which lies in the open interval  $]a, b[$ , since it divides the interval  $]a, b[$  internally in the ratio  $m : n$ . Hence the Rolle's theorem is verified.

**Ex. 6.** Verify Rolle's theorem for

$f(x) = x(x+3)e^{-x/2}$  in  $[-3, 0]$ .

**Sol.** We have  $f(x) = x(x+3)e^{-x/2}$ .

$\therefore f'(x) = (2x+3)e^{-x/2} + (x^2+3x)e^{-x/2} \cdot \left(-\frac{1}{2}\right)$   
 $= e^{-x/2} \left[ 2x+3 - \frac{1}{2}(x^2+3x) \right] = -\frac{1}{2}(x^2-x-6)e^{-x/2}$ ,

which exists for every value of  $x$  in the interval  $[-3, 0]$ . Hence  $f(x)$  is differentiable and so also continuous in the interval  $[-3, 0]$ . Also  $f(-3) = f(0) = 0$ .

Thus all the three conditions of Rolle's theorem are satisfied. So  $f'(x) = 0$  for at least one value of  $x$  lying in the open interval  $]-3, 0[$ .

Now  $f'(x) = 0 \Rightarrow -\frac{1}{2}(x^2-x-6)e^{-x/2} = 0$  or  $x^2-x-6 = 0$

$(x-3)(x+2) = 0$  or  $x = 3, -2$ .

Since the value  $x = -2$  lies in the open interval  $]-3, 0[$ , the Rolle's theorem is verified.

**Ex. 7.** If the functions  $f, g, h$  are continuous on  $[a, b]$  and twice differentiable on  $]a, b[$ , prove that there exist  $\xi, \eta \in ]a, b[$  such that

$$\begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix} = \frac{1}{2}(b-c)(c-a)(a-b) \begin{vmatrix} f(a) & f'(\xi) & f''(\eta) \\ g(a) & g'(\xi) & g''(\eta) \\ h(a) & h'(\xi) & h''(\eta) \end{vmatrix}$$

where  $a < c < b$ .

**Sol.** Consider the function  $F$  defined by

$$F(x) = \begin{vmatrix} f(a) & f(b) & f(x) \\ g(a) & g(b) & g(x) \\ h(a) & h(b) & h(x) \end{vmatrix} - \frac{(x-a)(x-b)}{(c-a)(c-b)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix}.$$

We have  $F(a) = F(b) = F(c) = 0$ . The function  $F$  satisfies the conditions of Rolle's theorem on  $[a, c]$  and  $[c, b]$ . Hence, we get

$F'(x_1) = 0 = F'(x_2)$  where  $a < x_1 < c$  and  $c < x_2 < b$ .

Again, applying Rolle's theorem for  $F''(x)$  on  $[x_1, x_2]$ , we get

$F''(\eta) = 0$ , where  $x_1 < \eta < x_2$ .

Now  $F''(\eta) = 0$

$$\Rightarrow \begin{vmatrix} f(a) & f(b) & f''(\eta) \\ g(a) & g(b) & g''(\eta) \\ h(a) & h(b) & h''(\eta) \end{vmatrix} = \frac{2}{(c-a)(c-b)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix} \quad \dots (1)$$

Again, let  $\phi$  be the function defined by

$$\phi(x) = \begin{vmatrix} f(a) & f(x) & f''(\eta) \\ g(a) & g(x) & g''(\eta) \\ h(a) & h(x) & h''(\eta) \end{vmatrix} - \frac{2(x-a)}{(c-a)(c-b)(b-a)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix}.$$

Then  $\phi(a) = 0$  and using (1), we have  $\phi(b) = 0$ .

Therefore applying Rolle's theorem for  $\phi$  on  $[a, b]$ , we have  $\phi'(\xi) = 0$  where  $a < \xi < b$ .

$$\begin{vmatrix} f(a) & f'(\xi) & f'(b) \\ g(a) & g'(\xi) & g'(b) \\ h(a) & h'(\xi) & h'(b) \end{vmatrix} = \frac{2}{(c-a)(c-b)(b-a)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix}$$

## EXERCISE 2

(Kanpur 2005)

- (i) State Rolle's theorem.  
(ii) Verify Rolle's theorem when  $f(x) = e^x \sin x$ ,  $a = 0$ ,  $b = \pi$ .
- Verify Rolle's theorem for the following functions:  
(i)  $f(x) = (x-4)^5(x-3)^4$  in the interval  $[3, 4]$ .  
(ii)  $f(x) = x^3 - 6x^2 + 11x - 6$ .  
(iii)  $f(x) = x^3 - 4x$  in  $[-2, 2]$ .  
(iv)  $f(x) = e^x(\sin x - \cos x)$  in  $[\pi/4, 5\pi/4]$ .  
(v)  $f(x) = 10x - x^2$  in  $[0, 10]$ .
- Discuss the applicability of Rolle's theorem to the function  $f(x) = x^2 + 1$ , when  $0 \leq x \leq 1$   
 $= 3 - x$ , when  $1 < x \leq 2$ .
- Show that between any two roots of  $e^x \cos x = 1$  there exists at least one root of  $e^x \sin x = 0$ .
- State and prove Rolle's theorem. Interpret it geometrically. Verify Rolle's theorem for the function  $f(x) = x^2$  in  $[-1, 1]$ .
- Verify the truth of Rolle's theorem for the function  $f(x) = x^2 - 3x + 2$  on the interval  $[1, 2]$ .
- Does the function  $f(x) = |x - 2|$  satisfy the conditions of Rolle's theorem in the interval  $[1, 3]$ . Justify your answer with correct reasoning.
- The function  $f$  is defined in  $[0, 1]$  as follows:  
$$f(x) = 1 \quad \text{for } 0 \leq x < \frac{1}{2}$$
$$= 2 \quad \text{for } \frac{1}{2} \leq x \leq 1.$$

Show that  $f(x)$  satisfies none of the conditions of Rolle's theorem, yet  $f'(x) = 0$  for many points in  $[0, 1]$ .

- If  $a + b + c = 0$ , then show that the quadratic equation  $3ax^2 + 2bx + c = 0$  has at least one root in  $]0, 1[$ .
- Let  $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n-1} + \dots + \frac{a_{n-1}}{2} + a_n = 0$ . Show that there exists at least one real  $x$  between 0 and 1 such that  $a_0 x^n + a_1 x^{n-1} + \dots + a_n = 0$ .
- If  $f(x) = \begin{vmatrix} \sin x & \sin \alpha & \sin \beta \\ \cos x & \cos \alpha & \cos \beta \\ \tan x & \tan \alpha & \tan \beta \end{vmatrix}$  where  $0 < \alpha < \beta < \pi/2$ , show that  $f'(\xi) = 0$ , where  $\alpha < \xi < \beta$ .

- Show that there is no real number  $k$  for which the equation  $x^3 - 3x + k = 0$ , has two distinct roots in  $[0, 1]$ .

## ANSWERS

- The given function is not differentiable at  $x = 1$  and so Rolle's theorem is not applicable to the given function in the interval  $[0, 2]$ .
- The function does not satisfy the third condition that  $f(x)$  must be differentiable in the open interval  $]1, 3[$ .

## § 6. LAGRANGE'S MEAN VALUE THEOREM.

**Theorem.** If a function  $f(x)$  is  
(i) continuous in a closed interval  $[a, b]$ ,  
and (ii) differentiable in the open interval  $]a, b[$  i.e.,  $a < x < b$ , then there exists at least one value ' $c$ ' of  $x$  lying in the open interval  $]a, b[$  such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

**Proof.** Consider the function  $\phi(x)$  defined by  $\phi(x) = f(x) + Ax$ , ... (1)  
where  $A$  is a constant to be chosen such that  $\phi(a) = \phi(b)$

$$\text{i.e., } f(a) + Aa = f(b) + Ab \quad \text{or} \quad A = -\frac{f(b) - f(a)}{b - a} \quad \dots (2)$$

(i) Now the function  $f$  is given to be continuous on  $[a, b]$  and the mapping  $x \rightarrow Ax$  is continuous on  $[a, b]$ , therefore  $\phi$  is continuous on  $[a, b]$ .

(ii) Also, since  $f$  is given to be differentiable on  $]a, b[$  and the mapping  $x \rightarrow Ax$  is differentiable on  $]a, b[$ , therefore,  $\phi$  is differentiable on  $]a, b[$ .

(iii) By our choice of  $A$ , we have  $\phi(a) = \phi(b)$ .  
From (i), (ii) and (iii), we find that  $\phi$  satisfies all the conditions of Rolle's theorem on  $[a, b]$ . Hence there exists at least one point, say  $x = c$ , of the open interval  $]a, b[$ , such that  $\phi'(c) = 0$ .

But  $\phi'(x) = f'(x) + A$ , from (1).

$$\therefore \phi'(c) = 0 \Rightarrow f'(c) + A = 0$$

$$\text{or } f'(c) = -A = \frac{f(b) - f(a)}{b - a}, \text{ from (2).}$$

This proves the theorem. It is usually known as the 'First Mean Value Theorem of Differential Calculus'.

## Another form of Lagrange's mean value theorem.

If in the above theorem, we take  $b = a + h$ , then a number  $c$ , lying between  $a$  and  $b$  can be written as  $c = a + \theta h$ , where  $\theta$  is some real number such that  $0 < \theta < 1$ .

Now Lagrange's theorem can be stated as follows:

If  $f$  be defined and continuous on  $[a, a + h]$  and differentiable on  $]a, a + h[$ , then there exists a point  $c = a + \theta h$  ( $0 < \theta < 1$ ) in the open interval  $]a, a + h[$  such that

$$\frac{f(a + h) - f(a)}{h} = f'(a + \theta h)$$

$$\text{or } f(a + h) - f(a) = hf'(a + \theta h).$$

### Geometrical interpretation of the mean value theorem.

Let  $y = f(x)$  and let  $ACB$  be the graph of  $y = f(x)$  in  $[a, b]$ . The coordinates of the point  $A$  are  $(a, f(a))$  and those of  $B$  are  $(b, f(b))$ . If the chord  $AB$  makes an angle  $\alpha$  with the  $x$ -axis, then

$$\tan \alpha = \frac{f(b) - f(a)}{b - a} = f'(c).$$

by Lagrange's mean value theorem where  $a < c < b$ .

Thus Lagrange's mean value theorem says that there is some point  $c$  in  $[a, b]$  such that the tangent to the curve at this point is parallel to the chord joining the points on the graph with abscissae  $a$  and  $b$ .

### § 7. IMPORTANT DEDUCTIONS FROM THE MEAN VALUE THEOREM.

**Theorem 1.** If a function  $f$  is continuous on  $[a, b]$ , differentiable on  $]a, b[$  and if  $f'(x) = 0$  for all  $x$  in  $]a, b[$ , then  $f(x)$  has a constant value throughout  $[a, b]$ .

**Proof.** Let  $c$  be any point of  $]a, b[$ . Then the function  $f$  is continuous on  $[a, c]$  and differentiable on  $]a, c[$ . Thus  $f$  satisfies all the conditions of Lagrange's mean value theorem on  $[a, c]$ . Consequently there exists a real number  $d$  between  $a$  and  $c$  i.e.,  $a < d < c$  such that

$$f(c) - f(a) = (c - a)f'(d).$$

But by hypothesis  $f'(x) = 0$  throughout the interval  $]a, b[$ , therefore, in particular  $f'(d) = 0$  and hence  $f(c) - f(a) = 0$  or  $f(c) = f(a)$ . Since  $c$  is any point of  $]a, b[$ , therefore, it gives that  $f(x) = f(a) \forall x$  in  $]a, b[$ . Thus  $f(x)$  has a constant value throughout  $[a, b]$ .

**Theorem 2.** If  $f(x)$  and  $\phi(x)$  are functions continuous on  $[a, b]$  and differentiable on  $]a, b[$  and if  $f'(x) = \phi'(x)$  throughout the interval  $]a, b[$ , then  $f(x)$  and  $\phi(x)$  differ only by a constant.

**Proof.** Consider the function  $F(x) = f(x) - \phi(x)$ . Throughout the interval  $]a, b[$ , we have

$$F'(x) = f'(x) - \phi'(x) = 0, \text{ because } f'(x) = \phi'(x).$$

Consequently, from theorem 1, we get

$$F(x) = \text{constant or } f(x) - \phi(x) = \text{constant}.$$

**Theorem 3.** If  $f'(x) = k$  for each point  $x$  of  $]a, b[$ ,  $k$  being a constant, then

$$f(x) = kx + C \quad \forall x \in [a, b], \text{ where } C \text{ is a constant}.$$

**Proof.** Consider the interval  $[a, x]$  such that  $[a, x]$  lies in the interval  $[a, b]$  i.e.,  $[a, x] \subset [a, b]$ . Since  $f'(x)$  exists  $\forall x \in [a, b]$ ,  $f$  is differentiable on  $[a, b]$  and hence on  $[a, x]$  and consequently continuous on  $[a, x]$ . Thus  $f$  satisfies all the conditions of Lagrange's mean value theorem on  $[a, x]$  and hence there is a point  $c \in ]a, x[$  such that

$$f(x) - f(a) = (x - a)f'(c).$$

But by hypothesis  $f'(x) = k \quad \forall x \in [a, b]$ , therefore, in particular  $f'(c) = k$  as  $a < c < x < b$  i.e.,  $a < c < b$ .

$$\text{Hence } f(x) - f(a) = (x - a)k \text{ or } f(x) = kx + f(a) - ak$$

$$f(x) = kx + C \text{ where } C = f(a) - ak \text{ is a constant.}$$

or

**Theorem 4.** If  $f$  is continuous on  $[a, b]$  and  $f'(x) \geq 0$  in  $]a, b[$ , then  $f$  is increasing in  $[a, b]$ .

**Proof.** Let  $x_1$  and  $x_2$  be any two distinct points of  $[a, b]$  such that  $x_1 < x_2$ . Then  $f$  satisfies the conditions of the Lagrange's mean value theorem in  $[x_1, x_2]$ . Consequently there exists a number  $c$  such that  $x_1 < c < x_2$ , and  $f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$ .

Now  $x_2 - x_1 > 0$  and  $f'(c) \geq 0$  (as  $f'(x) \geq 0 \quad \forall x \in ]a, b[$  and  $c$  is a point of  $]a, b[$ ), therefore

$$f(x_2) - f(x_1) \geq 0 \text{ i.e., } f(x_1) \leq f(x_2).$$

Thus  $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2) \quad \forall x_1, x_2 \in [a, b]$ .

Hence  $f$  is an increasing function in the interval  $[a, b]$ .

Similarly, we can prove that if  $f'(x) \leq 0$  in  $]a, b[$ , then  $f$  is decreasing in  $[a, b]$ .

**Corollary.** If  $f$  is continuous on  $[a, b]$ , then  $f$  is strictly increasing or strictly decreasing on  $[a, b]$  according as

$$f'(x) > 0 \text{ or } < 0 \text{ in } ]a, b[.$$

### § 8. CAUCHY'S MEAN VALUE THEOREM.

If two functions  $f(x)$  and  $g(x)$  are

- continuous in a closed interval  $[a, b]$ ,
- differentiable in the open interval  $]a, b[$ ,

and (iii)  $g'(x) \neq 0$  for any point of the open interval  $]a, b[$ , then there exists at least one value  $c$  of  $x$  in the open interval  $]a, b[$ , such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}, \quad a < c < b.$$

**Proof.** First we observe that as a consequence of condition (iii),  $g(b) - g(a) \neq 0$ . For if  $g(b) - g(a) = 0$  i.e.,  $g(b) = g(a)$ , then the function  $g(x)$  satisfies all the conditions of Rolle's theorem in  $[a, b]$  and consequently there is some  $x$  in  $]a, b[$  for which  $g'(x) = 0$ , thus contradicting the hypothesis that  $g'(x) \neq 0$  for any point of  $]a, b[$ .

Now consider the function  $F(x)$  defined on  $[a, b]$ , by setting

$$F(x) = f(x) + Ag(x), \quad \dots(1)$$

where  $A$  is a constant to be chosen such that  $F(a) = F(b)$

$$\text{i.e., } f(a) + Ag(a) = f(b) + Ag(b)$$

$$\text{or } -A = \frac{f(b) - f(a)}{g(b) - g(a)} \quad \dots(2)$$

Since  $g(b) - g(a) \neq 0$ , therefore  $A$  is a definite real number.

(i) Now  $f$  and  $g$  are continuous on  $[a, b]$ , therefore,  $F$  is also continuous on  $[a, b]$ .

(ii) Again, since  $f$  and  $g$  are differentiable on  $]a, b[$ , therefore  $F$  is also differentiable on  $]a, b[$ .

(iii) By our choice of  $A$ ,  $F(a) = F(b)$ .

Thus the function  $F(x)$  satisfies the conditions of Rolle's theorem in the interval  $[a, b]$ . Consequently there exists, at least one value, say  $c$ , of  $x$  in the open interval  $]a, b[$  such that  $F'(c) = 0$ .

But  $F'(x) = f'(x) + Ag'(x)$ , from (1).  
 $F'(c) = 0 \Rightarrow f'(c) + Ag'(c) = 0$   
 $\therefore -A = \frac{f'(c)}{g'(c)}$

From (2) and (3), we get  $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$

Another form. If  $b = a + h$ , then  $a + \theta h = a$  when  $\theta = 0$  and  $a + \theta h = b$  when  $\theta = 1$ . Therefore, if  $0 < \theta < 1$ , then  $a + \theta h$  means some value between  $a$  and  $b$ .  
 putting  $b = a + h$  and  $c = a + \theta h$ , the result of the above theorem can be written as  
 $\frac{f(a+h) - f(a)}{g(a+h) - g(a)} = \frac{f'(a+\theta h)}{g'(a+\theta h)}$ ,  $0 < \theta < 1$ .

**Note 1.** If we take  $g(x) = x$  for all  $x$  in  $[a, b]$ , then Cauchy's mean value theorem gives Lagrange's mean value theorem as a particular case. For  $g(x) = x$ ,  $g(b) = b$ ,  $g(a) = a$ ,  $g'(x) = 1$  and so  $g'(c) = 1$ . Putting these values in Cauchy's mean value theorem, we get Lagrange's mean value theorem. Thus Cauchy's mean value theorem is more general than Lagrange's mean value theorem.

**Note 2.** Cauchy's mean value theorem cannot be obtained by applying Lagrange's mean value theorem to the functions  $f$  and  $g$ .

For applying Lagrange's mean value theorem to  $f(x)$  and  $g(x)$  separately, we get  
 $f(b) - f(a) = (b-a)f'(c_1)$ , where  $a < c_1 < b$   
 $g(b) - g(a) = (b-a)g'(c_2)$ , where  $a < c_2 < b$ .

Dividing, we have  $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c_1)}{g'(c_2)}$

Note that here  $c_1$  is not necessarily equal to  $c_2$ .

### SOLVED EXAMPLES

**Ex. 1.** If  $f(x) = (x-1)(x-2)(x-3)$  and  $a = 0$ ,  $b = 4$ , find 'c' using Lagrange's mean value theorem.

**Sol.** We have

$$f(x) = (x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6.$$

$$\therefore f(a) = f(0) = -6 \text{ and } f(b) = f(4) = 6.$$

$$\therefore \frac{f(b) - f(a)}{b - a} = \frac{6 - (-6)}{4 - 0} = \frac{12}{4} = 3.$$

$$\text{Also } f'(x) = 3x^2 - 12x + 11 \text{ gives } f'(c) = 3c^2 - 12c + 11.$$

Putting these values in Lagrange's mean value theorem

$$\frac{f(b) - f(a)}{b - a} = f'(c), (a < c < b), \text{ we get}$$

$$3 = 3c^2 - 12c + 11 \text{ or } 3c^2 - 12c + 8 = 0$$

$$\text{or } c = \frac{12 \pm \sqrt{144 - 96}}{6} = 2 \pm \frac{2\sqrt{3}}{3}.$$

As both of these values of  $c$  lie in the open interval  $]0, 4[$ , hence both of these are the required values of  $c$ .

**Ex. 2.** Let  $f: [0, 1] \rightarrow \mathbb{R}$  be defined by

$$f(x) = (x-1)^2 + 2 \quad \forall x \in [0, 1].$$

Find the equation of the tangent to the graph of this curve which is parallel to the chord joining the points  $(0, 3)$  and  $(1, 2)$  of the curve.

**Sol.** Since  $f(x)$  is a polynomial function, therefore it is continuous on  $[0, 1]$  and differentiable in  $]0, 1[$ . Hence, by Lagrange's mean value theorem, there is some  $c \in ]0, 1[$  such that

$$\frac{f(1) - f(0)}{1 - 0} = f'(c) \text{ or } \frac{2 - 3}{1} = f'(c) \text{ or } -1 = f'(c).$$

$$\text{Now } f'(x) = 2(x-1) \text{ gives } f'(c) = 2(c-1).$$

$$\text{Thus } 2(c-1) = -1 \text{ i.e., } c = \frac{1}{2}.$$

$\therefore f(c) = \frac{9}{4}$ , so that the point of contact of the tangent is  $\left(\frac{1}{2}, \frac{9}{4}\right)$  and its slope is  $f'(c) = -1$ . Hence the equation of the required tangent is

$$y - \frac{9}{4} = -1 \left(x - \frac{1}{2}\right) \text{ or } 4x + 4y = 11.$$

**Ex. 3.** Compute the value of  $\theta$  in the first mean value theorem

$$f(x+h) = f(x) + hf'(x+\theta h), \text{ if } f(x) = ax^2 + bx + c.$$

**Sol.** Here  $f(x) = ax^2 + bx + c$ .

$$\therefore f(x+h) = a(x+h)^2 + b(x+h) + c,$$

$$f'(x) = 2ax + b, f'(x+\theta h) = 2a(x+\theta h) + b.$$

Substituting all these values in the Lagrange's mean value theorem, we get

$$a(x+h)^2 + b(x+h) + c = ax^2 + bx + c + h[2a(x+\theta h) + b] \quad \dots (1)$$

The relation (1) being identically true for all values of  $x$ , hence when  $x \rightarrow 0$ , we have

$$ah^2 + bh + c = c + h[2a\theta h + b]$$

$$\text{or } ah^2 = 2a\theta h^2 \text{ or } \theta = 1/2.$$

**Ex. 4.** A function  $f(x)$  is continuous in the closed interval  $[0, 1]$  and differentiable in the open interval  $]0, 1[$ , prove that

$$f'(x_1) = f(1) - f(0), \text{ where } 0 < x_1 < 1.$$

**Sol.** Here  $a = 0$ ,  $b = 1$  so that

$$\frac{f(b) - f(a)}{b - a} = \frac{f(1) - f(0)}{1 - 0} = f(1) - f(0).$$

If we take  $c = x_1$ , and substitute these values in the result of Lagrange's mean value theorem, we get

$$f(1) - f(0) = f'(x_1) \text{ where } 0 < x_1 < 1.$$

This is a particular case of Lagrange's mean value theorem. Students can give an independent proof of this.

**Ex. 5.** Separate the intervals in which the polynomial

$$2x^3 - 15x^2 + 36x + 1 \text{ is increasing or decreasing}$$

**Sol.** We have  $f(x) = 2x^3 - 15x^2 + 36x + 1$ .

$$f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3).$$

Now  $f'(x) > 0$  for  $x < 2$  or for  $x > 3$ ,  
 $f'(x) < 0$  for  $2 < x < 3$ , and  $f'(x) = 0$  for  $x = 2, 3$ .

Thus  $f'(x)$  is +ive in the intervals  $]-\infty, 2[$  and  $]3, \infty[$  and negative in the interval  $[2, 3]$ .

Hence  $f$  is monotonically increasing in the intervals  $]-\infty, 2[$ ,  $]3, \infty[$  and monotonically decreasing in the interval  $[2, 3]$ .

Ex. 6. Show that  $\frac{x}{1+x} < \log(1+x) < x$  for  $x > 0$ .

Sol. Let  $f(x) = \log(1+x) - \frac{x}{1+x}$ .  $\therefore f(0) = 0$ .

$$\text{Then } f'(x) = \frac{1}{1+x} - \frac{1 \cdot (1+x) - x \cdot 1}{(1+x)^2} = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2}$$

We observe that  $f'(x) > 0$  for  $x > 0$ . Hence  $f(x)$  is monotonically increasing in the interval  $[0, \infty[$ . Therefore

$$f(x) > f(0) \text{ for } x > 0 \text{ i.e., } \left[ \log(1+x) - \frac{x}{1+x} \right] > 0 \text{ for } x > 0$$

$$\text{i.e., } \log(1+x) > \frac{x}{1+x} \text{ for } x > 0. \quad \dots(1)$$

Again, let  $\phi(x) = x - \log(1+x)$ .  
 $\therefore \phi(0) = 0$ .

$$\text{Then } \phi'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x}$$

We observe that  $\phi'(x) > 0$  for  $x > 0$ . Hence  $\phi(x)$  is monotonically increasing in the interval  $[0, \infty[$ . Therefore

$$\phi(x) > \phi(0) \text{ for } x > 0 \text{ i.e., } [x - \log(1+x)] > 0 \text{ for } x > 0$$

$$\text{i.e., } x > \log(1+x) \text{ for } x > 0. \quad \dots(2)$$

From (1) and (2), we get

$$\frac{x}{1+x} < \log(1+x) < x \text{ for } x > 0.$$

Ex. 7. Verify Cauchy's mean value theorem for the functions  $x^2$  and  $x^3$  in the interval  $[1, 2]$ .

Sol. Let  $f(x) = x^2$  and  $g(x) = x^3$ . Then  $f(x)$  and  $g(x)$  are continuous in the closed interval  $[1, 2]$  and differentiable in the open interval  $]1, 2[$ . Also  $g'(x) = 3x^2 \neq 0$  for any point in the open interval  $]1, 2[$ . Hence by Cauchy's mean value theorem there exists at least one real number  $c$  in the open interval  $]1, 2[$ , such that

$$\frac{f(2) - f(1)}{g(2) - g(1)} = \frac{f'(c)}{g'(c)} \quad \dots(1)$$

$$\text{Now } \frac{f(2) - f(1)}{g(2) - g(1)} = \frac{4 - 1}{8 - 1} = \frac{3}{7}$$

$$\text{Also } f'(x) = 2x, g'(x) = 3x^2.$$

## DIFFERENTIABILITY

$$\therefore \frac{f'(c)}{g'(c)} = \frac{2c}{3c^2} = \frac{2}{3c} \quad \text{Putting these values in (1), we get } \frac{3}{7} = \frac{2}{3c} \text{ or } c = \frac{14}{9}$$

which lies in the open interval  $]1, 2[$ . Hence Cauchy's mean value theorem is verified.

Ex. 8. If in the Cauchy's mean value theorem, we write  $f(x) = e^x$  and  $g(x) = e^{-x}$ , show that 'c' is the arithmetic mean between  $a$  and  $b$ .

$$\text{Sol. Here } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{e^b - e^a}{e^{-b} - e^{-a}} = -e^a e^b = -e^{a+b},$$

$$\text{and } \frac{f'(x)}{g'(x)} = \frac{e^x}{-e^{-x}} \text{ so that } \frac{f'(c)}{g'(c)} = \frac{e^c}{-e^{-c}} = -e^{2c}.$$

Putting these values in Cauchy's mean value theorem, we get

$$-e^{a+b} = -e^{2c} \text{ or } 2c = a + b \text{ or } c = \frac{1}{2}(a + b).$$

Thus  $c$  is the arithmetic mean between  $a$  and  $b$ .

Ex. 9. If in the Cauchy's mean value theorem, we write

(i)  $f(x) = \sqrt{x}$  and  $g(x) = 1/\sqrt{x}$ , then  $c$  is the geometric mean between  $a$  and  $b$ ,

and if (ii)  $f(x) = 1/x^2$  and  $g(x) = 1/x$ , then  $c$  is the harmonic mean between  $a$  and  $b$ .

(Bundelkhand 2005)

$$\text{Sol. (i) Here } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{\sqrt{b} - \sqrt{a}}{(1/\sqrt{b}) - (1/\sqrt{a})} = -\sqrt{ab},$$

$$\text{and } \frac{f'(x)}{g'(x)} = \frac{\frac{1}{2}x^{-1/2}}{-\frac{1}{2}x^{-3/2}} \text{ so that } \frac{f'(c)}{g'(c)} = -\frac{c^{-1/2}}{c^{-3/2}} = -c.$$

Putting these values in Cauchy's mean value theorem, we get

$$-\sqrt{ab} = -c \text{ or } c = \sqrt{ab}$$

i.e.,  $c$  is the geometric mean between  $a$  and  $b$ .

$$\text{(ii) Here } \frac{f(b) - f(a)}{g(b) - g(a)} = \frac{(1/b^2) - (1/a^2)}{(1/b) - (1/a)} = \frac{a+b}{ab}$$

$$\text{and } \frac{f'(x)}{g'(x)} = \frac{-2x^{-3}}{-x^{-2}} \text{ so that } \frac{f'(c)}{g'(c)} = \frac{-2c^{-3}}{-c^{-2}} = \frac{2}{c}.$$

Putting these values in Cauchy's mean value theorem, we get

$$\frac{a+b}{ab} = \frac{2}{c} \text{ or } c = \frac{2ab}{a+b}$$

i.e.,  $c$  is the harmonic mean between  $a$  and  $b$ .

## EXERCISE 3

1. State Lagrange's mean value theorem. Test if Lagrange's mean value theorem holds for the function  $f(x) = |x|$  in the interval  $[-1, 1]$ . (Kanpur 2010)
2. If  $f(x) = 1/x$  in  $[-1, 1]$ , will the Lagrange's mean value theorem be applicable to  $f(x)$ ?
3. Verify Lagrange's mean value theorem for the function  $f: [-1, 1] \rightarrow \mathbb{R}$  given by  $f(x) = x^3$ .

4. Find  $\xi$  of the mean value theorem, if  $f(x) = x(x-1)(x-2)$ ;  $a = 0$ ,  $b = \frac{1}{2}$ .
5. Find  $\xi$  of mean value theorem when  
 (i)  $f(x) = x^3 - 3x - 2$  in  $[-2, 3]$   
 (ii)  $f(x) = 2x^2 + 3x + 4$  in  $[1, 2]$   
 (iii)  $f(x) = x(x-1)$  in  $[1, 2]$   
 (iv)  $f(x) = x^2 - 3x - 1$  in  $[-\frac{11}{7}, \frac{13}{7}]$
6. Show that any chord of the parabola  $y = Ax^2 + Bx + C$  is parallel to the tangent at the point whose abscissa is same as that of the middle point of the chord.
7. If  $f''(\xi)$  exists for all points in  $[a, b]$  and  $\frac{f(c) - f(a)}{c - a} = \frac{f(b) - f(c)}{b - c}$ , where  $a < c < b$ , then there is a number  $\xi$  such that  $a < \xi < b$  and  $f''(\xi) = 0$ .
8. State the conditions for the validity of the formula  $f(x+h) = f(x) + hf'(x + \theta h)$  and investigate how far these conditions are satisfied and whether the result is true, when  $f(x) = x \sin(1/x)$  (being defined to be zero at  $x = 0$ ) and  $x < 0 < x + h$ .
9. (a) Show that  $x^3 - 3x^2 + 3x + 2$  is monotonically increasing in every interval.  
 (b) Show that  $\log(1+x) - \frac{2x}{2+x}$  is increasing when  $x > 0$ .
10. Determine the intervals in which the function  $(x^4 + 6x^3 + 17x^2 + 32x + 32)e^{-x}$  is increasing or decreasing.
11. Use the function  $f(x) = x^{1/x}$ ,  $x > 0$  to determine the bigger of the two numbers  $e^\pi$  and  $\pi^e$ .
12. Show that the set of all  $x$  for which  $\log(1+x) \leq x$  is equal to  $[0, \infty[$ .
13. Use Lagrange's mean value theorem to prove that  $1 + x < e^x < 1 + xe^x$ ,  $\forall x > 0$ .
14. Let  $\mathbb{R}$  be the set of real numbers and  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that for all  $x$  and  $y$  in  $\mathbb{R}$ ,  $|x - y|^3$ . Prove that  $f(x)$  is a constant function.
15. Show that  $x^n - a = 0$  has at most one real positive root if  $n$  is a positive integer.
16. If  $f(0) = 0$  and  $f''(x)$  exists in  $[0, \infty[$ , show that  $f'(x) - \frac{f(x)}{x} = \frac{1}{2}xf''(\xi)$  where  $0 < \xi < x$  and deduce that if  $f''(x) > 0$  for  $x > 0$ , then  $\frac{f(x)}{x}$  is strictly increasing in  $[0, \infty[$ .
17. If  $f''(x) \geq 0$ ,  $\forall x \in [a, b]$ , show that  $x_1, x_2 \in [a, b] \Rightarrow f\left(\frac{x_1 + x_2}{2}\right) \leq \frac{f(x_1) + f(x_2)}{2}$ .
18. Let  $f$  be continuous on  $[a-h, a+h]$  and differentiable on  $]a-h, a+h[$ . Prove that there is a real number  $\theta$  between 0 and 1 such that  $f(a+h) - 2f(a) + f(a-h) = h[f'(a+\theta h) - f'(a-\theta h)]$ .
19. Show that the function  $f'$  if it exists in an interval, cannot have an ordinary or removable discontinuity in that interval.
20. If  $a = -1$ ,  $b \geq 1$  and  $f(x) = 1/|x|$ , show that the conditions of Lagrange's mean value theorem are not satisfied in the interval  $[a, b]$ , but the conclusion of the theorem is true if and only if  $b > 1 + \sqrt{2}$ .

21. State Cauchy's mean value theorem. Verify Cauchy's mean value theorem for  $f(x) = \sin x$ ,  $g(x) = \cos x$  in  $[-\pi/2, 0]$ . (Kanpur 2007)
22. If  $f(x) = x^2$ ,  $g(x) = \cos x$ , then find the point  $c \in ]0, \pi/2[$  which gives the result of Cauchy's mean value theorem in the interval  $[0, \pi/2]$  for the functions  $f(x)$  and  $g(x)$ .
23. Use Cauchy's mean value theorem to show that  $\frac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha} = \cot \theta$ , where  $0 < \alpha < \theta < \beta < \frac{\pi}{2}$ .
24. Use Cauchy's mean value theorem to evaluate  $\lim_{x \rightarrow 1} \frac{\cos \frac{1}{x} \pi x}{\log(1/x)}$ .
25. If  $f(x)$ ,  $g(x)$  and  $h(x)$  are functions such that  
 (i) all the three are continuous on  $[a, b]$   
 (ii) all the three are differentiable in  $]a, b[$ , then  $\begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(b) & g(b) & h(b) \\ f(a) & g(a) & h(a) \end{vmatrix} = 0$ , where  $a < c < b$ .
26. If  $f''(x)$  exists for all  $x$  in  $[a, b]$ , show that  $f(c) - f(a) \frac{b-c}{b-a} - f(b) \frac{c-a}{b-a} - \frac{1}{2}(c-a)(c-b)f''(\xi) = 0$  where  $c$  and  $\xi$  both lie in  $]a, b[$ .
27. If  $f''$  is continuous on  $[a, b]$  and differentiable on  $]a, b[$ , then prove that  $f(b) - f(a) - (b-a)[f'(a) + f'(b)] = -\frac{(b-a)^3}{12}f'''(d)$ , where  $d$  is a real number lying between  $a$  and  $b$ .
28. If  $\phi(x) = f(x) + f(1-x)$  and  $f''(x) < 0$  for all  $x \in [0, 1]$ , show that  $\phi$  increases in  $[0, 1/2]$  and decreases in  $[1/2, 1]$ . Hence or otherwise prove that  $\pi < \frac{\sin \pi x}{x(1-x)} \leq 4$  when  $0 < x < 1$ .

# ANSWERS

- The mean value theorem does not hold since the given function is not differentiable at  $x = 0$ .
- not applicable.
- $1 - \frac{\sqrt{21}}{6}$
- (i)  $\pm \sqrt{7/3}$ , (ii)  $3/2$ , (iii)  $3/2$ , (iv)  $1/7$ .
- Non-differentiability is not satisfied in  $x < 0 < x + h$  since  $f(x)$  is non-differentiable at  $x = 0$ .
- Increasing in the intervals  $[-2, -1]$  and  $[0, 1]$  and decreasing in the intervals  $[-\infty, -2]$ ,  $[-1, 0]$  and  $[1, \infty[$ .
- $e^\pi$  is bigger than  $\pi^e$ .
- Root of the equation  $\sin c - (8c/\pi^2) = 0$  in the open interval  $]\pi/6, \pi/2[$ .
- $\pi/2$ .

## Multiple Choice Questions.

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

7. If  $y = \log x$ , then  $D^n \log x$  is  
 (a)  $\frac{(-1)^n (n-1)!}{x^n}$  (b)  $\frac{(-1)^{n-1} (n-1)!}{x^n}$   
 (c)  $\frac{(-1)^{n-1} n!}{x^n}$  (d)  $\frac{(-1)^{n-1} (n-1)!}{x^{n+1}}$
8. If  $x = a(\cos \theta + \theta \sin \theta)$ ,  $y = a(\sin \theta - \theta \cos \theta)$ , then  $\frac{d^2 y}{dx^2}$  is  
 (a)  $\frac{1}{a} \sec^3 \theta$  (b)  $a \sec^3 \theta$  (c)  $\frac{\theta \sec^3 \theta}{a}$  (d)  $a \theta \sec^3 \theta$
9. By Leibnitz's theorem we find the  $n^{\text{th}}$  differential coefficient of the ..... of two functions.  
 (a) sum (b) difference (c) product (d) quotient
10. If  $y = e^{\tan^{-1} x}$ , then  $(1+x^2)y_{n+2} + [2(n+1)x-1]y_{n+1} = \dots$   
 (a)  $-n(n-1)y_n$  (b)  $\frac{n}{2}(n+1)y_n$   
 (c)  $-\frac{n}{2}(n-1)y_n$  (d)  $-n(n+1)y_n$

## True or False.

Write 'T' for true and 'F' for false statement.

11. If  $y = f(x)$ , then the  $n^{\text{th}}$  differential coefficient of  $y_r$  is the  $(n+r)^{\text{th}}$  differential coefficient of  $y$ .
12. If  $y = e^{ax} \sin(bx+c)$ , then  
 $D^n \{e^{ax} \sin(bx+c)\} = r^n e^{ax} \sin\{bx+c+(n+1)\phi\}$ ,  
 where  $r = (a^2 + b^2)^{1/2}$  and  $\phi = \tan^{-1}(b/a)$ .
13. While applying Leibnitz's theorem if we observe that one of the two functions is such that all its differential coefficients after a certain stage become zero, then we should take that function as second function.

## ANSWERS

1.  $a^n \sin\left(ax+b+\frac{n\pi}{2}\right)$  2.  $(-1)^n n! a^n (ax+b)^{-n-1}$   
 3.  $\frac{1}{2}[e^x - (5)^{n/2} e^x \cos(2x+n \tan^{-1} 2)]$  4.  $-y$  5.  $n^2 y_{n-1}$   
 6.  $(-1)^n e^{-x}$  7. (b). 8. (a). 9. (c).  
 10. (d). 11. T. 12. F. 13. T.

## 5 EXPANSIONS OF FUNCTIONS

## § 1. ACCURATE STATEMENT OF TAYLOR'S THEOREM. (Without proof).

If  $f(x)$  is a single-valued function of  $x$  such that  
 (i) all the derivatives of  $f(x)$  upto  $(n-1)^{\text{th}}$  are continuous in  $a \leq x \leq a+h$ ,  
 and (ii)  $f^{(n)}(x)$  exists in  $a < x < a+h$ , then

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{h^n}{n!} f^{(n)}(a+\theta h), \text{ where } 0 < \theta < 1.$$

## Taylor's series.

Suppose  $f(x)$  possesses continuous derivatives of all orders in the interval  $[a, a+h]$ . Then for every positive integral value of  $n$ , we have

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + R_n, \dots (1)$$

where

$$R_n = \frac{h^n}{n!} f^{(n)}(a+\theta h), (0 < \theta < 1).$$

Suppose  $R_n \rightarrow 0$ , as  $n \rightarrow \infty$ . Then taking limits of both sides of (1) when  $n \rightarrow \infty$ , we get

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^n}{n!} f^{(n)}(a) + \dots (2)$$

The series given in (2) is known as Taylor's infinite series for the expansion of  $f(a+h)$  as a power series in  $h$ .

## § 2. MACLAURIN'S SERIES.

Suppose  $f(x)$  possesses continuous derivatives of all orders in the interval  $[0, x]$ . Then for every positive integral value of  $n$ , we have

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + R_n, \dots (1)$$

where

$$R_n = \frac{x^n}{n!} f^{(n)}(\theta x), (0 < \theta < 1).$$

Suppose  $R_n \rightarrow 0$ , as  $n \rightarrow \infty$ . Then taking limits of both sides of (1) when  $n \rightarrow \infty$ , we get

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots (2)$$

The series given in (2) is known as Maclaurin's infinite series for the expansion of  $f(x)$  as a power series in  $x$ . Maclaurin's series is a particular case of Taylor's series. If in Taylor's series we put  $a = 0$  and  $h = x$ , we get Maclaurin's series.

Maclaurin's expansion of  $f(x)$  fails if any of the functions  $f(x), f'(x), f''(x), \dots$  becomes infinite or discontinuous at any point of the interval  $[0, x]$  or if  $R_n$  does not tend to zero as  $n \rightarrow \infty$ .

### § 3. FORMAL EXPANSIONS OF FUNCTIONS.

We have seen that for the validity of the expansion of a function  $f(x)$  as an infinite Maclaurin's series, it is necessary that  $R_n \rightarrow 0$  as  $n \rightarrow \infty$ . But to examine the behaviour of  $R_n$  as  $n \rightarrow \infty$  is not an easy job because in many cases it is not possible to find a general expression for the  $n$ th derivative of the function to be expanded. So in this chapter we shall simply obtain formal expansion of a function  $f(x)$  without showing that  $R_n \rightarrow 0$  as  $n \rightarrow \infty$ . Such an expansion will not give us any idea of the range of values of  $x$  for which the expansion is valid. To obtain such an expansion of  $f(x)$  we have only to calculate the values of its derivatives for  $x = 0$  and substitute them in the infinite Maclaurin's series

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

For the convenience of the students we shall now give formal proofs of Maclaurin's and Taylor's theorems without bothering about the nature of  $R_n$  as  $n \rightarrow \infty$ .

#### Maclaurin's Theorem.

Let  $f(x)$  be a function of  $x$  which possesses continuous derivatives of all orders in the interval  $[0, x]$ . Assuming that  $f(x)$  can be expanded as an infinite power series in  $x$ , we have

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

**Proof.** Suppose  $f(x) = A_0 + A_1 x + A_2 x^2 + A_3 x^3 + \dots$  ... (1)

Let the expansion (1) be differentiable term by term any number of times. Then by successive differentiation, we have

$$f'(x) = A_1 + 2A_2 x + 3A_3 x^2 + 4A_4 x^3 + \dots$$

$$f''(x) = 2 \cdot 1 A_2 + 3 \cdot 2 A_3 x + 4 \cdot 3 A_4 x^2 + \dots$$

$$f'''(x) = 3 \cdot 2 \cdot 1 A_3 + 4 \cdot 3 \cdot 2 A_4 x + \dots \text{ and so on.}$$

Putting  $x = 0$  in each of these relations, we get

$$f(0) = A_0, f'(0) = A_1, f''(0) = 2! A_2, f'''(0) = 3! A_3, \dots$$

Substituting these values of  $A_0, A_1, A_2, \dots$  in (1), we get

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

This is Maclaurin's Theorem. If we denote  $f(x)$  by  $y$ , then Maclaurin's theorem can also be written in the following way:

$$y = (y)_0 + \frac{x}{1!} (y_1)_0 + \frac{x^2}{2!} (y_2)_0 + \frac{x^3}{3!} (y_3)_0 + \dots + \frac{x^n}{n!} (y_n)_0 + \dots$$

#### Taylor's theorem.

(Bundelkhand 2005)

Let  $f(x)$  be a function of  $x$  which possesses continuous derivatives of all orders in the interval  $[a, a+h]$ . Assuming that  $f(a+h)$  can be expanded as an infinite power series in  $h$ , we have

$$f(a+h) = f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^n}{n!} f^{(n)}(a) + \dots$$

### EXPANSIONS OF FUNCTIONS

**Proof.** Suppose  $f(a+h) = A_0 + A_1 h + A_2 h^2 + A_3 h^3 + \dots$

Let the expansion (1) be differentiable term by term any number of times w.r.t.  $h$ . Then by successive differentiation w.r.t.  $h$ , we have

$$f'(a+h) = A_1 + 2A_2 h + 3A_3 h^2 + \dots$$

$$f''(a+h) = 2 \cdot 1 A_2 + 3 \cdot 2 A_3 h + \dots$$

$$f'''(a+h) = 3 \cdot 2 \cdot 1 A_3 + \dots \text{ and so on.}$$

Putting  $h = 0$  in each of the above relations, we get

$$f(a) = A_0, f'(a) = A_1, f''(a) = 2! A_2, f'''(a) = 3! A_3, \text{ and so on.}$$

$\therefore A_0 = f(a), A_1 = f'(a), A_2 = \frac{1}{2!} f''(a), A_3 = \frac{1}{3!} f'''(a), \text{ and so on.}$

Substituting these values of  $A_0, A_1, A_2, A_3, \dots$  in (1), we get

$$f(a+h) = f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^n}{n!} f^{(n)}(a) + \dots$$

This is Taylor's theorem. Another useful form is obtained on replacing  $h$  by  $(x-a)$ . Thus

$$f(x) = f(a) + (x-a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^n}{n!} f^{(n)}(a) + \dots$$

which is an expansion of  $f(x)$  as a power series in  $(x-a)$ .

**Note.** If we expand  $f(x+h)$ , by Taylor's theorem, as a power series in  $h$ , then the result is as follows:

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \dots + \frac{h^n}{n!} f^{(n)}(x) + \dots$$

### SOLVED EXAMPLES

**Ex. 1.** Expand  $e^x$  in ascending powers of  $x$ .

(Bundelkhand 2008)

**Sol.** Let  $f(x) = e^x$ . Then  $f(0) = 1, f^{(n)}(x) = e^x$  so that  $f^{(n)}(0) = 1$ , where  $n = 1, 2, 3, 4, \dots$

Substituting these values in Maclaurin's series

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots, \text{ we get}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^n}{n!} + \dots$$

This is known as Exponential series.

**Ex. 2.** Expand  $(1+x)^n$  in ascending powers of  $x$ .

**Sol.** Let  $f(x) = (1+x)^n$ , so that  $f(0) = 1$ .

We have  $f^{(m)}(x) = n(n-1)\dots(n-m+1)(1+x)^{n-m}$ .

$\therefore f^{(m)}(0) = n(n-1)\dots(n-m+1)$ .

Putting  $m = 1, 2, 3, \dots$ , we have

$$f'(0) = n, f''(0) = n(n-1), f'''(0) = n(n-1)(n-2), \text{ and so on.}$$

Substituting these values in Maclaurin's series for  $f(x)$ , we get

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-m+1)}{m!}x^m + \dots$$

This is known as Binomial series. If  $n$  is a positive integer, the series will terminate at  $(n+1)$  terms.

Ex. 3. Expand  $\cos x$ .

Sol. Let  $f(x) = \cos x$ . Then  $f(0) = 1$

$$f'(x) = -\sin x, \quad f'(0) = 0$$

$$f''(x) = -\cos x, \quad f''(0) = -1$$

$$f'''(x) = \sin x, \quad f'''(0) = 0$$

In general  $f^{(n)}(x) = \cos(x + \frac{1}{2}n\pi)$  so that

$$f^{(n)}(0) = \cos \frac{1}{2}n\pi = 0 \text{ when } n = 2m \text{ and } = (-1)^m \text{ when } n = 2m+1$$

Hence substituting these values in Maclaurin's series, we get

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^m \frac{x^{2m+1}}{(2m+1)!} + \dots$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^m \frac{x^{2m+1}}{(2m+1)!} + \dots$$

This is known as Sine series.

Similarly we may obtain Cosine series:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^m \frac{x^{2m}}{(2m)!} + \dots$$

Ex. 4. Expand  $\log(1+x)$  by Maclaurin's theorem. (Kanpur 2000)

Sol. Let  $f(x) = \log(1+x)$ . Then  $f(0) = \log 1 = 0$ ,  $f^{(n)}(x) = \frac{(-1)^{n-1}}{(x+1)^n}$

so that  $f^{(n)}(0) = (-1)^{n-1} (n-1)!$  where  $n = 1, 2, 3, 4, \dots$

Now by Maclaurin's theorem,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$$

Substituting the values of  $f(0), f'(0), f''(0), \dots$ , we get

$$\log(1+x) = 0 + x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots$$

$$\dots + \frac{x^n}{n!} (-1)^{n-1} (n-1)! + \dots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n-1} \frac{x^n}{n} + \dots$$

Ex. 5. Apply Maclaurin's theorem to find the expansion in ascending powers of  $x$  of  $\log_e(1+e^x)$  to the terms containing  $x^4$ .

Sol. Let  $y = \log_e(1+e^x)$ . Then  $(y)_0 = \log_e(1+e^0) = \log_e 2$ .

Now  $y_1 = \frac{e^x}{1+e^x} = \frac{(1+e^x) - 1}{1+e^x} = 1 - \frac{1}{1+e^x}$  so that  $(y_1)_0 = 1 - \frac{1}{2} = \frac{1}{2}$ .

$$y_2 = 0 + \frac{e^x}{(1+e^x)^2} = \frac{e^x}{1+e^x} \cdot \frac{1}{1+e^x} = y_1(1-y_1) = y_1 - y_1^2$$

we that

$$(y_2)_0 = (y_1)_0 - [(y_1)_0]^2 = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$y_3 = y_2 - 2y_1 y_2 \text{ so that } (y_3)_0 = \frac{1}{4} - 2 \cdot \frac{1}{2} \cdot \frac{1}{4} = 0$$

$$y_4 = y_3 - 2y_1 y_3 \text{ so that } (y_4)_0 = 0 - 2 \cdot \left(\frac{1}{2}\right)^2 = -\frac{1}{2} \text{ and so on.}$$

Now by Maclaurin's theorem, we have

$$y = (y)_0 + x(y_1)_0 + \frac{x^2}{2!}(y_2)_0 + \frac{x^3}{3!}(y_3)_0 + \frac{x^4}{4!}(y_4)_0 + \dots$$

$$\therefore \log(1+e^x) = \log 2 + x \cdot \frac{1}{2} + \frac{x^2}{2!} \cdot \frac{1}{4} + \frac{x^3}{3!} \cdot 0 + \frac{x^4}{4!} \cdot \left(-\frac{1}{2}\right) + \dots$$

$$= \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$$

Ex. 6. Expand  $\log\{x + \sqrt{1+x^2}\}$  in ascending powers of  $x$  and find the general term.

Sol. Let  $y = \log\{x + \sqrt{1+x^2}\}$ .

$$\text{Then } y_1 = \frac{1}{x + \sqrt{1+x^2}} \cdot \left[1 + \frac{2x}{2\sqrt{1+x^2}}\right] = \frac{1}{\sqrt{1+x^2}} \quad \dots(1)$$

$$\therefore y_1^2(1+x^2) - 1 = 0. \quad \dots(2)$$

Differentiating again, we get  $(1+x^2)2y_1 y_2 + 2xy_1^2 = 0$

$$2y_1[(1+x^2)y_2 + xy_1] = 0$$

$$\text{or } (1+x^2)y_2 + xy_1 = 0$$

$$\text{or } 2y_1 \neq 0$$

$$\therefore \dots(3)$$

Now differentiating (3)  $n$  times by Leibnitz's theorem, we get

$$(1+x^2)y_{n+2} + 2 + n \cdot y_{n+1} \cdot 2x + \frac{n(n-1)}{1 \cdot 2} y_n \cdot 2 + y_{n+1} \cdot x + n \cdot y_n \cdot 1 = 0$$

$$\text{or } (1+x^2)y_{n+2} + 2 + (2n+1)xy_{n+1} + n^2 y_n = 0$$

Putting  $x = 0$  in (1), (2), (3) and (4), we get

$$(y)_0 = 0, (y_1)_0 = 1, (y_2)_0 = 0$$

$$\text{and } (y_{n+2})_0 = -n^2 (y_n)_0 \quad \dots(5)$$

Now putting  $n = 1, 3, 5, \dots$  in (5), we get

$$(y_3)_0 = -1^2 (y_1)_0 = -1^2$$

$$(y_5)_0 = (-3^2) (y_3)_0 = (-3^2) (-1^2) = 3^2 \cdot 1^2$$

$$(y_7)_0 = (-5^2) (y_5)_0 = (-5^2) (-3^2) (-1^2) = -5^2 \cdot 3^2 \cdot 1^2 \text{ and so on.}$$

Putting  $n-2$  in place of  $n$  in (5), we get

$$(y_n)_0 = \{-(n-2)^2\} (y_{n-2})_0 \quad \dots(6)$$

$$= \{-(n-2)^2\} \{-(n-4)^2\} (y_{n-4})_0$$

$$\therefore \text{replacing } n \text{ by } n-2 \text{ in (6), we have } (y_{n-2})_0 = -(n-4)^2 (y_{n-4})_0$$

Thus if  $n$  is odd, we have

$$(y_n)_0 = \{-(n-2)^2\} \{-(n-4)^2\} \dots \{(-5)^2\} \{(-3)^2\} \{(-1)^2\} \cdot 1$$

$$= (-1)^{(n-1)/2} (n-2)^2 (n-4)^2 \dots 5^2 \cdot 3^2 \cdot 1^2.$$

Again, putting  $n = 2, 4, 6, \dots$  in (5), we get

$$(y_2)_0 = -2^2, (y_4)_0 = 0, (y_6)_0 = -4^2, (y_8)_0 = 0, \text{ and so on.}$$

Thus, if  $n$  is even, we have  $(y_n)_0 = 0$ .

Now by Maclaurin's theorem, we have

$$y = (y)_0 + \frac{x}{1!} (y_1)_0 + \frac{x^2}{2!} (y_2)_0 + \frac{x^3}{3!} (y_3)_0 + \dots$$

$$\therefore \log \{x + \sqrt{1+x^2}\} = 0 + x \cdot 1 + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!} \cdot (-1^2) + \frac{x^4}{4!} \cdot 0$$

$$+ \frac{x^5}{5!} (3^2 \cdot 1^2) + \frac{x^6}{6!} \cdot 0 + \frac{x^7}{7!} (-5^2 \cdot 3^2 \cdot 1^2) + \dots$$

$$= x - \frac{x^3}{3!} \cdot 1^2 + \frac{x^5}{5!} (3^2 \cdot 1^2) - \frac{x^7}{7!} (5^2 \cdot 3^2 \cdot 1^2) + \dots$$

The general term =  $\frac{x^n}{n!} (y_n)_0$ , where  $(y_n)_0$  is given by (7) when  $n$  is odd and  $(y_n)_0 = 0$ , when  $n$  is even.

Putting  $2n-1$  in place of  $n$  in (7), we find that

$$(y_{2n-1})_0 = (-1)^{n-1} (2n-3)^2 (2n-5)^2 \dots 5^2 \cdot 3^2 \cdot 1^2.$$

Hence  $\log \{x + \sqrt{1+x^2}\}$

$$= x - \frac{x^3}{3!} \cdot 1^2 + \frac{x^5}{5!} \cdot 3^2 \cdot 1^2 - \frac{x^7}{7!} \cdot 5^2 \cdot 3^2 \cdot 1^2 + \dots$$

$$+ (-1)^{n-1} 1^2 \cdot 3^2 \cdot 5^2 \dots (2n-3)^2 \cdot \frac{x^{2n-1}}{(2n-1)!} + \dots$$

Ex 7. If  $y = \sin^{-1} x = a_0 + a_1 x + a_2 x^2 + \dots$ , prove that

$$(n+1)(n+2)a_{n+2} = n^2 a_n \quad (\text{Meerut 2010B; Kumaun 06})$$

Sol. Let  $y = \sin^{-1} x$ .

$$\text{Then } y_1 = \frac{1}{\sqrt{1-x^2}} \quad \dots(1)$$

$$\therefore y_1^2 (1-x^2) - 1 = 0. \quad \dots(2)$$

Differentiating again, we get  $(1-x^2) 2y_1 y_2 - 2xy_1^2 = 0$

$$\text{or } 2y_1 [(1-x^2)y_2 - xy_1] = 0 \quad \dots(3)$$

$$\text{or } (1-x^2)y_2 - xy_1 = 0,$$

$$\text{since } 2y_1 \neq 0.$$

Now differentiating (3)  $n$  times by Leibnitz's theorem, we get

$$(1-x^2)y_{n+2} + n \cdot y_{n+1} \cdot (-2x) + \frac{n(n-1)}{1 \cdot 2} y_n (-2) - y_{n+1} \cdot x - n y_n \cdot 1 = 0$$

Putting  $x = 0$  in (4), we get  $(y_{n+2})_0 = n^2 (y_n)_0$

By Maclaurin's theorem, we have

$$y = (y)_0 + \frac{x}{1!} (y_1)_0 + \frac{x^2}{2!} (y_2)_0 + \frac{x^3}{3!} (y_3)_0 + \dots + \frac{x^n}{n!} (y_n)_0 + \dots \quad \dots(4)$$

Also we are given that

$$y = \sin^{-1} x = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots \quad \dots(5)$$

Equating the coefficients of  $x^n$  in the two expansions for  $y$ , we get  $a_n = \frac{(y_n)_0}{n!}$

$$\therefore \frac{a_{n+2}}{a_n} = \frac{(y_{n+2})_0}{(n+2)!} \cdot \frac{n!}{(y_n)_0} = \frac{(y_{n+2})_0}{(y_n)_0} \cdot \frac{1}{(n+2)(n+1)}$$

$$= \frac{n^2}{(n+2)(n+1)}, \text{ substituting for } \frac{(y_{n+2})_0}{(y_n)_0} \text{ from (5).}$$

$$\text{Hence } (n+1)(n+2)a_{n+2} = n^2 a_n.$$

Ex. 8. Expand  $\sin x$  in powers of  $\left(x - \frac{1}{2}\pi\right)$  by using Taylor's series.

(Meerut 2005; Rohilkhand 06; Agra 06)

Sol. Let  $f(x) = \sin x$ . We want to expand  $f(x)$  in powers of  $x - \frac{1}{2}\pi$ .

We can write  $f(x) = f\left[\frac{1}{2}\pi + \left(x - \frac{1}{2}\pi\right)\right]$ .

Now expanding  $f\left[\frac{1}{2}\pi + \left(x - \frac{1}{2}\pi\right)\right]$  by Taylor's theorem in powers of  $\left(x - \frac{1}{2}\pi\right)$ , we get

$$f(x) = f\left[\frac{1}{2}\pi + \left(x - \frac{1}{2}\pi\right)\right] = f\left(\frac{1}{2}\pi\right) + \left(x - \frac{1}{2}\pi\right)f'\left(\frac{1}{2}\pi\right)$$

$$+ \frac{1}{2!} \left(x - \frac{1}{2}\pi\right)^2 f''\left(\frac{1}{2}\pi\right) + \frac{1}{3!} \left(x - \frac{1}{2}\pi\right)^3 f'''\left(\frac{1}{2}\pi\right) + \dots \quad \dots(1)$$

Now  $f(x) = \sin x$ . Therefore  $f\left(\frac{1}{2}\pi\right) = \sin \frac{1}{2}\pi = 1$ .

$$f'(x) = \cos x \text{ giving } f'\left(\frac{1}{2}\pi\right) = \cos \frac{1}{2}\pi = 0,$$

$$f''(x) = -\sin x \text{ so that } f''\left(\frac{1}{2}\pi\right) = -\sin \frac{1}{2}\pi = -1,$$

$$f'''(x) = -\cos x \text{ so that } f'''\left(\frac{1}{2}\pi\right) = -\cos \frac{1}{2}\pi = 0,$$

$$f^{iv}(x) = \sin x \text{ so that } f^{iv}\left(\frac{1}{2}\pi\right) = \sin \frac{1}{2}\pi = 1, \text{ etc.}$$

Substituting these values in (1), we get

$$\sin x = 1 + \left(x - \frac{1}{2}\pi\right) \cdot 0 + \frac{1}{2!} \left(x - \frac{1}{2}\pi\right)^2 \cdot (-1) + \frac{1}{3!} \left(x - \frac{1}{2}\pi\right)^3 \cdot 0 + \frac{1}{4!} \left(x - \frac{1}{2}\pi\right)^4 \cdot (-1) + \dots$$

$$= 1 - \frac{1}{2!} \left(x - \frac{1}{2}\pi\right)^2 + \frac{1}{4!} \left(x - \frac{1}{2}\pi\right)^4 - \dots$$

Ex. 9. Expand  $\log \sin(x+h)$  in powers of  $h$  by Taylor's theorem. (Meerut 2010; Bundelkhand 07)

Sol. Let  $f(x+h) = \log \sin(x+h)$ .  
Then by Taylor's theorem, we have

$$\log \sin(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots$$

Now  $f(x+h) = \log \sin(x+h)$ .  
 $\therefore f(x) = \log \sin x$ ,  
 $f'(x) = (1/\sin x) \cdot \cos x = \cot x$ ,  
 $f''(x) = -\operatorname{cosec}^2 x$ ,  
 $f'''(x) = 2 \operatorname{cosec} x \operatorname{cosec} x \cot x = 2 \operatorname{cosec}^2 x \cot x$ .

Hence,  
 $\log \sin(x+h) = \log \sin x + h \cot x - \frac{1}{2} h^2 \operatorname{cosec}^2 x + \frac{1}{3} h^3 \operatorname{cosec}^2 x \cot x + \dots$

Ex. 10. Use Taylor's theorem to prove that  
 $\tan^{-1}(x+h) = \tan^{-1} x + h \sin \theta \frac{\sin \theta}{1} - (h \sin \theta)^2 \frac{\sin 2\theta}{2} + (h \sin \theta)^3 \frac{\sin 3\theta}{3} + \dots + (-1)^{n-1} (h \sin \theta)^n \frac{\sin n\theta}{n} + \dots$   
(Agra 2007; Gorakhpur 06)

where  $\theta = \cot^{-1} x$ .

Sol. Let  $y = f(x) = \tan^{-1} x$ .

Then  $y_1 = \frac{1}{1+x^2} = \frac{1}{(x+i)(x-i)} = \frac{1}{2i} \left[ \frac{1}{x-i} - \frac{1}{x+i} \right]$

or  $y_1 = \frac{1}{2i} [(x-i)^{-1} - (x+i)^{-1}]$  ...[1]

Differentiating (1),  $(n-1)$  times, we get

$$y_n = \frac{1}{2i} [(-1)^{n-1} (n-1)! (x-i)^{-n} - (-1)^{n-1} (n-1)! (x+i)^{-n}]$$

or  $y_n = \frac{(-1)^{n-1} (n-1)!}{2i} [(x-i)^{-n} - (x+i)^{-n}]$  ...[2]

Now put  $x = r \cos \theta$ ,  $1 = r \sin \theta$  in (2). Then

$$y_n = \frac{(-1)^{n-1} (n-1)!}{2i} \cdot r^{-n} [(\cos \theta - i \sin \theta)^{-n} - (\cos \theta + i \sin \theta)^{-n}]$$

$$= \frac{(-1)^{n-1} (n-1)!}{2i} r^{-n} [(\cos n\theta + i \sin n\theta) - (\cos n\theta - i \sin n\theta)],$$

by De Moivre's theorem

$$= \frac{(-1)^{n-1} (n-1)!}{2i} r^{-n} \cdot 2i \sin n\theta$$

$$= (-1)^{n-1} (n-1)! \sin^n \theta \sin n\theta.$$

Hence  $f^{(n)}(x) = (-1)^{n-1} (n-1)! \sin^n \theta \sin n\theta$ ,  
where  $\cot \theta = x$ , i.e.,  $\theta = \cot^{-1} x$ . [ $\because r^{-1} = 1/r = \sin \theta$ ]

Putting  $n = 1, 2, 3, \dots$ , we get

$$f'(x) = \sin \theta \cdot \sin \theta, f''(x) = -\sin^2 \theta \sin 2\theta,$$

$$f'''(x) = 2! \sin^3 \theta \sin 3\theta, \text{ and so on.}$$

Substituting these values in Taylor's series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots + \frac{h^n}{n!} f^{(n)}(x) + \dots, \text{ we get}$$

$$\tan^{-1}(x+h) = \tan^{-1} x + h \sin \theta \cdot \sin \theta - \frac{h^2}{2!} \sin^2 \theta \sin 2\theta$$

$$+ \frac{h^3}{3!} \sin^3 \theta \sin 3\theta - \dots + \frac{h^n}{n!} (-1)^{n-1} (n-1)! \sin^n \theta \sin n\theta + \dots$$

or  $\tan^{-1}(x+h) = \tan^{-1} x + h \sin \theta \cdot \frac{\sin \theta}{1} - (h \sin \theta)^2 \frac{\sin 2\theta}{2} + (h \sin \theta)^3 \frac{\sin 3\theta}{3} + \dots + (-1)^{n-1} (h \sin \theta)^n \frac{\sin n\theta}{n} + \dots$

### EXERCISE 1

- (i) State Maclaurin's theorem. (Meerut 2000; Bundelkhand 01, 08, 11; Agra 07)
- (i) State Taylor's theorem. (Bundelkhand 2006, 08, 11)
- Write the expansion of  $a^x$  by Maclaurin's theorem.
- Expand the following functions by Maclaurin's theorem :
 

|                                       |                     |
|---------------------------------------|---------------------|
| 3. $\tan x$ .                         | 4. $e^{x \cos x}$ . |
| 5. $\tan^{-1} x$ . (Bundelkhand 2001) | 6. $\sec x$ .       |
- Use Maclaurin's formula to show that  $e^x \sec x = 1 + x + \frac{2x^2}{2!} + \frac{4x^3}{3!} + \dots$  (Meerut 2004)
- Obtain by Maclaurin's theorem the first five terms in the expansion of  $e^{\sin x}$ . (Bundelkhand 2007)
- Expand by Maclaurin's theorem  $\frac{e^x}{1+e^x}$  as far as the term  $x^3$ . (Meerut 2006B)
- Obtain by Maclaurin's theorem the first five terms in the expansion of  $\log(1+\sin x)$ . (Meerut 2007)

11. Find the first three terms in the expansion in the powers of  $x$  of  $\log(1 + \tan x)$ .
12. Apply Maclaurin's theorem to prove that  $\log \sec x = \frac{1}{2}x^2 + \frac{1}{12}x^4 + \frac{1}{45}x^6 + \dots$  (Bundelkhand 2011)

13. Show that

$$(i) e^x \cos x = 1 + x - \frac{2x^3}{3!} - \frac{2x^4}{4!} - \frac{2x^5}{5!} + \frac{2x^7}{7!} + \dots$$

$$+ \cos\left(\frac{1}{4}n\pi\right) \cdot \frac{2^{n/2}}{n!} x^n + \dots$$

$$(ii) e^x \sin x = x + x^2 + \frac{2}{3!}x^3 - \frac{2^2}{5!}x^5 - \dots + \sin\left(\frac{1}{4}n\pi\right) \frac{2^{n/2}}{n!} x^n + \dots$$

(Meerut 2003; Gorakhpur 06)

14. Expand  $\sinh x \cos x$  to fifth powers of  $x$ .
15. Apply Maclaurin's theorem to prove that

$$(i) e^{ax} \sin bx = bx + abx^2 + \frac{3a^2b - b^3}{3!}x^3 + \dots$$

$$+ \frac{(a^2 + b^2)^{n/2}}{n!} x^n \sin\left(n \tan^{-1} \frac{b}{a}\right) + \dots$$

$$(ii) e^{ax} \cos bx = 1 + ax + \frac{a^2 - b^2}{2}x^2 + \frac{a(a^2 - 3b^2)}{3!}x^3 + \dots$$

$$+ \frac{(a^2 + b^2)^{n/2}}{n!} x^n \cos\left(n \tan^{-1} \frac{b}{a}\right) + \dots$$

16. Show that  $e^x \cos \alpha \cos(x \sin \alpha) = 1 + x \cos \alpha + \frac{x^2}{2!} \cos 2\alpha + \frac{x^3}{3!} \cos 3\alpha + \dots$  (Rohilkhand 2007)

17. (i) Expand  $\sin^{-1}(x+h)$  in powers of  $x$  as far as the term  $x^3$ .  
[Hint. Use Taylor's series]

$$(ii) \text{ Prove that } \log(x+h) = \log h + \frac{x}{h} - \frac{x^2}{2h^2} + \frac{x^3}{3h^3} - \dots$$

18. Expand  $\tan^{-1} x$  in powers of  $\left(x - \frac{1}{4}\pi\right)$ .

[Hint. Let  $f(x) = \tan^{-1} x$ . We can write  $f(x) = f\left[\frac{1}{4}\pi + \left(x - \frac{1}{4}\pi\right)\right]$ . Now apply Taylor's theorem]

19. Expand  $\sin\left(\frac{1}{4}\pi + \theta\right)$  in powers of  $\theta$ .

20. (i) Expand  $2x^3 + 7x^2 + x - 1$  in powers of  $x - 2$ . (Meerut 2004B, 05B; Gorakhpur 06)

(ii) Write the value of  $\alpha$ , if by Taylor's theorem  $2x^3 + 7x^2 + x - 1 = \alpha + 53(x - 2) + 19(x - 2)^2 + 2(x - 2)^3$ . (Meerut 2001)

21. Expand  $\log \sin x$  in powers of  $(x - a)$ .

- (i) If  $y = e^{a \sin^{-1} x}$ , show that  $(1 - x^2)y_n + 2 - (2n + 1)xy_{n+1} - (n^2 + a^2)y_n = 0$ . (Meerut 2001, 06)

Hence by Maclaurin's theorem, show that

$$e^{a \sin^{-1} x} = 1 + ax + \frac{a^2}{2!}x^2 + \frac{a(12 + a^2)}{3!}x^3 + \dots$$

Also deduce that  $e^\theta = 1 + \sin \theta + \frac{1}{2!}\sin^2 \theta + \frac{2}{3!}\sin^3 \theta + \dots$  (Kumaun 2008)

- (ii) If  $y = \sin(m \sin^{-1} x)$ , then show that

$$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + m^2 y = 0.$$

Hence or otherwise expand  $\sin m\theta$  in powers of  $\sin \theta$ .

23. If  $y = e^{\tan^{-1} x}$ , prove that  $(1 + x^2)y_n + 2 + \{2(n + 1)x - 1\}y_{n+1} + n(n + 1)y_n = 0$ .

Hence or otherwise, find out the coefficient of  $x^5$  if  $e^{\tan^{-1} x}$  is expanded in powers of  $x$ .

24. Expand  $(\sin^{-1} x)^2$  in ascending powers of  $x$ .

$$\text{Deduce that } \theta^2 = 2 \cdot \frac{\sin^2 \theta}{2!} + 2^2 \cdot \frac{2 \sin^4 \theta}{4!} + 2^2 \cdot 4^2 \cdot \frac{2 \sin^6 \theta}{6!} + \dots$$

25. If  $y = \sin \log(x^2 + 2x + 1)$ , prove that  $(x + 1)^2 y_n + 2 + (2n + 1)(x + 1)y_{n+1} + (n^2 + 4)y_n = 0$ .

Hence or otherwise expand  $y$  in ascending powers of  $x$  as far as  $x^6$ .

26. By Maclaurin's theorem or otherwise find the expansion of  $y = \sin(e^x - 1)$  upto and including the term in  $x^4$ . Find also the first two non-vanishing terms in the expansion of  $x$  as a series of ascending powers of  $y$ .

27. Expand  $\log\{1 - \log(1 - x)\}$  in powers of  $x$  by Maclaurin's theorem as far as the term  $x^3$ .

By substituting  $\frac{x}{1+x}$  for  $x$  deduce the expansion of  $\log\{1 + \log(1 + x)\}$  as far as the term in  $x^3$ .

28. If  $y = \sin^{-1} x / \sqrt{1 - x^2}$  when  $-1 < x < 1$ , and  $-\frac{1}{2}\pi < \sin^{-1} x < \frac{1}{2}\pi$ , prove that

$$(1 - x^2) \frac{d^{n+1} y}{dx^{n+1}} - (2n + 1)x \frac{d^n y}{dx^n} - n^2 \frac{d^{n-1} y}{dx^{n-1}} = 0. \quad (\text{Meerut 2007B})$$

Assuming that  $y$  can be expanded in ascending powers of  $x$  in the form

$$a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$$

prove that  $(n + 1)a_{n+1} = na_{n-1}$ , and hence obtain the general term of the expansion.

29. If  $y = e^{m \tan^{-1} x} = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$ , prove that  $(n + 1)a_{n+1} + (n - 1)a_{n-1} = ma_n$ .

D-126

30. If  $e^x = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$ , prove that

$$a_{n+1} = \frac{1}{n+1} \left\{ a_n + \frac{a_{n-1}}{1!} + \frac{a_{n-2}}{2!} + \dots + \frac{a_{n-r}}{r!} + \dots + \frac{a_0}{n!} \right\}$$

and hence calculate the first six terms in the expansion.

[Hint. Let  $y = e^x$ . Then  $y_1 = ye^x$ .

$$y_{n+1} = y_n e^x + {}^nC_1 y_{n-1} e^x + {}^nC_2 y_{n-2} e^x + \dots + {}^nC_n y e^x.$$

Now put  $x = 0$  on both sides and note that  $\frac{(y_n)_0}{n!} = a_n$ .

31. Prove by Maclaurin's theorem that

$$[x + \sqrt{1+x^2}]^m = 1 + mx + \frac{m^2}{2!} x^2 + \frac{m(m^2-1^2)}{3!} x^3 + \frac{m^2(m^2-2^2)}{4!} x^4 + \dots$$

32. Prove that

$$f(mx) = f(x) + (m-1)x f'(x) + \frac{(m-1)^2}{2!} x^2 f''(x) + \frac{(m-1)^3}{3!} x^3 f'''(x) + \dots$$

33. Prove that

$$(i) f\left(\frac{x^2}{1+x}\right) = f(x) - \frac{x}{1+x} f'(x) + \frac{x^2}{(1+x)^2} \frac{f''(x)}{2!} - \dots$$

$$(ii) f(x) = f(0) + x f'(x) - \frac{x^2}{2!} f''(x) + \frac{x^3}{3!} f'''(x) - \dots$$

[Hint. (i) Write  $f\left(\frac{x^2}{1+x}\right) = f\left(x - \frac{x}{1+x}\right)$ .

Now apply Taylor's theorem.

(ii) We have  $f(0) = f(x-x)$ .

Apply Taylor's theorem and transpose the terms to get the result.]

### ANSWERS

2.  $1 + x \log a + \frac{(x \log a)^2}{2!} + \dots + \frac{(x \log a)^n}{n!} + \dots$

3.  $x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots$

4.  $1 + x + \frac{x^2}{2} - \frac{x^3}{3} - \frac{11x^4}{24} - \frac{x^5}{5} + \dots$

5.  $x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)} + \dots$

6.  $1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \dots$

9.  $\frac{1}{2} + \frac{1}{4}x - \frac{x^2}{8} + \frac{1}{3!} + \dots$

11.  $x - \frac{1}{2}x^2 + \frac{2}{3}x^3 + \dots$

8.  $1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \dots$

10.  $x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{24} - \dots$

14.  $x - \frac{2x^3}{3!} - \frac{4x^5}{5!} + \dots$

### EXPANSIONS OF FUNCTIONS

D-127

17.  $\sin^{-1} h + x(1-h^2)^{-1/2} + \frac{x^2}{2!} h(1-h^2)^{-3/2} + \dots$

18.  $\tan^{-1}(\pi/4) + \left(x - \frac{1}{4}\pi\right) / (1 + \pi^2/16) - \pi \left(x - \frac{1}{4}\pi\right)^2 / \{4(1 + \pi^2/16)^2\} + \dots$

19.  $\frac{1}{\sqrt{2}} \left(1 + \frac{\theta^2}{2!} - \frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{\theta^5}{5!} - \dots\right)$

20. (i)  $45 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3$

(ii) 45.

21.  $\log \sin a + (x-a) \cot a - \frac{(x-a)^2}{2!} \operatorname{cosec}^2 a + \frac{(x-a)^3}{3!} 2 \operatorname{cosec}^2 a \cot a + \dots$

22. (ii)  $\sin m\theta = m \sin \theta + \frac{m(1^2-m^2)}{3!} \sin^3 \theta + \frac{m(1^2-m^2)(3^2-m^2)}{5!} \sin^5 \theta + \dots$

23.  $1/24$ .

24.  $2 \cdot \frac{x^2}{2!} + \frac{2 \cdot 2^2}{4!} x^4 + \frac{2 \cdot 2^2 \cdot 4^2}{6!} x^6 + \dots + \frac{2 \cdot 2^2 \cdot 4^2 \dots (2n-2)^2}{(2n)!} x^{2n} + \dots$

25.  $y = 2x - x^2 - \frac{2}{3}x^3 + \frac{3}{2}x^4 - \frac{5}{3}x^5 + \frac{3}{2}x^6 + \dots$

26.  $x + \frac{x^2}{2!} - \frac{5x^4}{24} + \dots, y = \frac{y^2}{2} + \dots$

27.  $x + \frac{x^3}{6} + \dots, x - x^2 + \frac{7x^3}{6} + \dots$

28.  $a_{2m} = 0, a_{2m+1} = \frac{2m(2m-2)(2m-4)\dots 2}{(2m+1)(2m-1)\dots 3}$

30.  $e' = e + ex + ex^2 + \frac{5e}{6}x^3 + \frac{5e}{8}x^4 + \frac{13e}{30}x^5 + \dots$

### EXERCISE 2 (Objective Questions)

#### Fill In The Blanks.

Fill in the blanks "...", so that the following statements are complete and correct.

1. By Maclaurin's theorem expansion of  $\sin^{-1}x$  is .....

2. By Maclaurin's theorem

$$y = \log(\sec x + \tan x) = x + \frac{x^3}{6} + \frac{x^5}{24} + \dots, \text{ then } (y_3)_0 = \dots$$

3. By Maclaurin's theorem

$$y = e^x \sin x = x + x^2 + \frac{2}{3!}x^3 + \dots + \frac{2^{n/2} \sin(n\pi/4)}{n!} x^n + \dots$$

then  $(y_3)_0 = \dots$

(Meerut 2001)

4. By Taylor's theorem the expansion of  $\log(x+h)$  in ascending powers of  $x$  is .....

Sol. Let  $y = \lim_{x \rightarrow 0} \left( \frac{\tan x}{x} \right)^{1/x^2}$

$$\therefore \log y = \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left( \frac{\tan x}{x} \right) = \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left[ \frac{1}{x} \left( x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right) \right]$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left[ 1 + \left( \frac{x^2}{3} + \frac{2}{15}x^4 + \dots \right) \right]$$

$$= \lim_{x \rightarrow 0} \frac{\left( \frac{x^2}{3} + \frac{2}{15}x^4 + \dots \right) - \frac{1}{2} \left( \frac{x^2}{3} + \frac{2}{15}x^4 + \dots \right)^2 + \dots}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x^2}{3} + \text{terms containing higher powers of } x}{x^2}$$

$$= \lim_{x \rightarrow 0} \left[ \frac{1}{3} + \text{terms containing powers of } x \text{ only in the numerator} \right]$$

$$= \frac{1}{3}$$

$$\therefore y = e^{1/3}$$

## EXERCISE 2

Evaluate the following limits :

- $\lim_{x \rightarrow \infty} \frac{x}{e^x}$
- $\lim_{x \rightarrow \infty} \frac{e^x}{x^3}$
- $\lim_{x \rightarrow 1} \frac{\log(1-x)}{\cot \pi x}$
- $\lim_{x \rightarrow \infty} \frac{\log x}{a^x}, a > 1$
- $\lim_{x \rightarrow a} \frac{\log(x-a)}{\log(e^x - e^a)}$
- $\lim_{x \rightarrow \frac{1}{2}} \frac{\sec \pi x}{\tan 3\pi x}$
- $\lim_{x \rightarrow \infty} \left\{ \frac{(\log x)^3}{x} \right\}$
- $\lim_{x \rightarrow \infty} \frac{x(\log x)^3}{1+x+x^2}$
- $\lim_{x \rightarrow \infty} x \tan \frac{1}{x}$
- $\lim_{x \rightarrow 0} \frac{\log \tan 2x}{\log \tan 3x}$

(Bundelkhand 2001)

## INDETERMINATE FORMS

- $\lim_{x \rightarrow \pi/2} \frac{\log \left( x - \frac{1}{2} \pi \right)}{\tan x}$
- $\lim_{x \rightarrow a} \frac{c \{ e^{1/(x-a)} - 1 \}}{\{ e^{1/(x-a)} + 1 \}}$
- $\lim_{x \rightarrow 1} \left\{ \frac{2}{x^2 - 1} - \frac{1}{x - 1} \right\}$
- $\lim_{x \rightarrow \pi/2} (\sec x - \tan x)$
- $\lim_{x \rightarrow 1} \left( \frac{1}{\log x} - \frac{x}{\log x} \right)$
- $\lim_{x \rightarrow 0} \left( \frac{\cot x - \frac{1}{x}}{x} \right)$
- $\lim_{x \rightarrow 0} \left( \frac{\operatorname{cosec} x - \cot x}{x} \right)$
- $\lim_{x \rightarrow 0} \left( \frac{1}{e^x - 1} - \frac{1}{x} \right)$
- $\lim_{x \rightarrow \pi/2} \left( x \tan x - \frac{\pi}{2} \sec x \right)$
- $\lim_{x \rightarrow 0} x \log x$
- $\lim_{x \rightarrow 0} \sin x \log x$
- $\lim_{x \rightarrow \infty} x(a^{1/x} - 1)$
- $\lim_{x \rightarrow \infty} 2^x \sin \frac{a}{2^x}$
- $\lim_{x \rightarrow \pi/2} (1 - \sin x) \tan x$
- $\lim_{x \rightarrow 0} x^m (\log x)^n$ , where  $m$  and  $n$  are positive integers.
- $\lim_{x \rightarrow 0} x^x$
- $\lim_{x \rightarrow 0} (\cos x)^{1/x}$
- $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$
- $\lim_{x \rightarrow \pi/2} (\sin x)^{\tan x}$
- $\lim_{x \rightarrow \pi/2} (\sec x)^{\cot x}$
- $\lim_{x \rightarrow \pi/4} (\tan x)^{\tan 2x}$

(Garhwal 2002)

(Agra 2003)

(Agra 2002)

(Garhwal 2003)

32.  $\lim_{x \rightarrow a} \left(2 - \frac{x}{a}\right)^{\tan(\pi x/2a)}$   
 33.  $\lim_{x \rightarrow 1} (1 - x^2)^{1/\log(1-x)}$   
 34.  $\lim_{x \rightarrow 1} x^{1/(1-x)}$   
 35.  $\lim_{x \rightarrow \infty} (a_0 x^m + a_1 x^{m-1} + \dots + a_m)^{1/x}$   
 36.  $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x}\right)^{1/x}$   
 37.  $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x}\right)^{1/x^3}$   
 38.  $\lim_{x \rightarrow 0} \left(\frac{\sinh x}{x}\right)^{1/x^2}$   
 39.  $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x}\right)^{1/x^2}$   
 40.  $\lim_{x \rightarrow 0} \left\{ \frac{2(\cosh x - 1)}{x^2} \right\}^{1/x^2}$   
 41.  $\lim_{x \rightarrow \infty} \left\{ \frac{\log x}{x} \right\}^{1/x}$   
 42.  $\lim_{x \rightarrow \infty} \left( \frac{\pi}{2} - \tan^{-1} x \right)^{1/x}$   
 43.  $\lim_{x \rightarrow 0} (\operatorname{cosec} x)^{1/\log x}$   
 44.  $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}$

## ANSWERS

- |                      |                     |                      |
|----------------------|---------------------|----------------------|
| 1. 0.                | 2. $\infty$ .       | 3. 0.                |
| 4. 0.                | 5. 1.               | 6. 3.                |
| 7. 0.                | 8. 0.               | 9. 1.                |
| 10. 1.               | 11. 0.              | 12. c.               |
| 13. $-\frac{1}{2}$ . | 14. 0.              | 15. -1.              |
| 16. $-\frac{1}{3}$ . | 17. $\frac{1}{2}$ . | 18. $-\frac{1}{2}$ . |
| 19. -1.              | 20. 0.              | 21. 0.               |
| 22. $\log a$ .       | 23. a.              | 24. 0.               |
| 25. 0.               | 26. 1.              | 27. 1.               |
| 28. $e^{-1/2}$ .     | 29. 1.              | 30. 1.               |
| 31. $1/e$ .          | 32. $e^{2/\pi}$ .   | 33. e.               |
| 34. $1/e$ .          | 35. 1.              | 36. 1.               |
| 37. $\infty$ .       | 38. $e^{1/6}$ .     | 39. $e^{-1/6}$ .     |

(Garhwal 2001, 2003)

(Kumaun 2008)

(Kumaun 2003)

EXERCISE 3  
(Objective Questions)

## Fill In The Blanks.

Fill in the blanks "...", so that the following statements are complete and correct.

- $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \dots$
- $\lim_{x \rightarrow 0} \frac{\log(1+kx^2)}{1-\cos x} = \dots$
- $\lim_{x \rightarrow \infty} \frac{x^2+2x}{5-3x^2} = \dots$
- $\lim_{x \rightarrow 1} \left(\sec \frac{\pi}{2x}\right) \cdot \log x = \dots$
- The value of  $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3} = \dots$
- The value of  $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x} = \dots$

(Agra 2002)

(Agra 2003)

## Multiple Choice Questions.

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The value of  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2}$  is  
 (a) 0 (b)  $\frac{2}{3}$   
 (c)  $\frac{1}{6}$  (d) 1
- Which of the following is not an indeterminate form?  
 (a)  $\frac{\infty}{\infty}$  (b)  $0 \times \infty$   
 (c)  $1^0$  (d)  $0^0$
- The value of the  $\lim_{x \rightarrow 0} \frac{\log \tan x}{\log x}$  is  
 (a) 0 (b) 1  
 (c) -1 (d) None of these
- Which of the following is an indeterminate form?  
 (a)  $\infty + \infty$  (b)  $\infty \times \infty$   
 (c)  $1^\infty$  (d)  $0^\infty$

(Garhwal 2002)